MAHATMA GANDHI UNIVERSITY

UNDER GRADUATE VOCATIONAL

PROGRAMMES (HONOURS) SYLLABUS

MGU-B.VOC. (Honours)

(2025 Admission Onwards)

Faculty: Physical Education and Sport Sciences

Expert Committee: Service Excellence and Hospitality

Programme: B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training

Mahatma Gandhi University
Priyadarshini Hills
Kottayam – 686560, Kerala, India

Contents

Preface

Welcome to the curriculum for the Bachelor of Vocational Studies in Advanced Multi Sports and Fitness Training (Honours), with a Minor in Fitness Training. This specialized four-year undergraduate program is meticulously designed to equip students with the professional knowledge, practical competencies, and leadership skills essential for thriving in the dynamic world of multi-sport coaching, athlete development, and fitness science.

The **Major in Multi Sports and Fitness Training** offers a broad and in-depth exploration of physical education, applied sports sciences, and performance training across multiple disciplines. Students will engage with foundational and advanced topics including anatomy and physiology, biomechanics, exercise science, sports nutrition, athletic conditioning, motor development, sports psychology, and coaching pedagogy. With a strong emphasis on applied learning, the program blends theory with extensive practical sessions, fieldwork, and internships to prepare students for real-world challenges in sports training and athlete performance enhancement.

In addition, the programme enables students to develop focused expertise in the science and practice of personal and group fitness instruction. Covering areas such as exercise prescription, functional training, corrective techniques, health screening, fitness assessment, and lifestyle coaching, this component supports students in becoming effective fitness trainers capable of designing inclusive and impactful programs for diverse populations.

This curriculum integrates academic rigor with hands-on learning, offering students opportunities to train, teach, coach, and innovate in various sporting and fitness environments. With pathways leading to careers in sports coaching, athletic training, health and wellness, performance analysis, and fitness entrepreneurship, graduates will be well-prepared for both employment and self-employment in the growing fitness and sports industry.

Students are encouraged to embrace a mindset of exploration, critical thinking, and professional integrity throughout their academic journey. By actively participating in classroom sessions, practical labs, internships, and community engagement projects, they will not only develop subject mastery but also the confidence and adaptability required in contemporary fitness and sports careers.

We extend our best wishes for your academic and professional growth and look forward to witnessing your contributions as dedicated and innovative leaders in the field of multi-sport and fitness training.

EXPERT COMMITTEE FOR SERVICE EXCELLENCE AND HOSPITALITY

1	Rekha K Nair (Convener)	Associate Professor of Business Administration	NSS College, Rajakumari, Idukki
2	Praveen Thariyan	Associate Professor of Physical Education	St.Dominic's College, Kanjirappally
3	Dr. Bindu M	Associate Professor of Physical Education & HoD	UC College, Aluva
4	Aneetta Elizabeth Manuel	Assistant Professor (B.Voc)	Henry Baker College, Melukavu
5	Dinu Dennis	Assistant Professor (B.Voc)	St. Paul's College, Kalamassery
6	Anish Thanikkal	B.Voc Nodal Officer	SSV College, Valayanchirangara
7	Ani Ealias	B.Voc Nodal Officer	Mar Thoma College for Women, Perumbavoor
8	Devika S	Assistant Professor (B.Voc)	St. Xaviers College for Women, Aluva
9	Remya P M	B.Voc Nodal Officer	Alphonsa College, Pala
10	Soumya T K	B.Voc Nodal Officer	Nirmala College, Muvattupuzha
11	Surya N S	Assistant Professor of Tourism Studies	Sree Narayana Arts & Science College, Kumarakom,Kottayam
12	Binuraj C R	Assistant Professor of Business Administration	SAS SNDP Yogam College, Konni
13	Dr. John K Babu	Assistant Professor of Business Administration	Baselius College, Kottayam
14	Shreekanth S K	Assistant Professor of Physical Education	Maharaja's College, Ernakulam
15	Hardin Rixon	B.Voc Nodal Officer	St. Paul's College, Kalamassery

SYLLABUS WORKSHOP PARTICIPANTS

1	Haneefa K G	Assistant Professor of Physical Education & HoD	MES College Marampally
2	Rajesh Kovvammal	Assistant Professor & HoD, Multi Sports and Fitness Training	MES College Marampally
3	Joshna E V	Assistant Professor, Multi Sports and Fitness Training	MES College Marampally
4	Harshad K H	Instructor, Multi Sports and Fitness Training	MES College Marampally

SYLLABUS INDEX

Name of the Major Subject: Advanced Course in Multi Sports and Fitness Training

Semester: 1

Course Code	Title of the Course	Type of the Course	Credit	Hours /week	Hour Distribution /week		
					L	P	O
MG1SDCMFT100	Foundation of Resistance & Functional Training	SDC	4	5	3	2	0
MG1SDCMFT101	Principles of Health & Fitness	SDC	4	4	4	0	0
MG1SDCMFT102	Basics of Practical Yoga and Healing	SDC	4	5	3	2	0
MG1MDCMFT100	First aid and Emergency Care	MDC	3	3	3	0	0
MG1SDCMFT103	On-the Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Semester: 2

Course Code	Title of the Course	Type of the Course	Credit	Hours /week	Hour Distribution /week		
					L	P	O
MG2SDCMFT100	Human Body: Structure, function & Energy Metabolism	SDC	4	4	4	0	0
MG2SDCMFT101	Fitness Recovery: Massage, Wellness, and Recreation	SDC	4	5	3	2	0
MG2SDCMFT102	Martial Arts for Fitness & Self-defense - I - Taekwondo, Krav Maga	SDC	4	5	3	2	0
MG2MDCMFT100	Self Defense for Fitness	MDC	3	4	2	2	0
MG2SDCMFT103	On-the-Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Semester 3

Course Code	Title of the Course	Type of the Course	Credit	Hours /week	Hour Distribution /week		
					L	P	O
MG3SDCMFT200	Foundation of Personal Training and Exercise Programming	SDC	4	4	4	0	0
MG3SDCMFT201	Applied Aerobic fitness and Rhythmic workouts	SDC	4	5	3	2	0
MG3SDCMFT202	Fitness Assessment & Evaluation	SDC	4	5	3	2	0
MG3MPCMFT200	Self Defense Training and Fitness	MPC	4	5	3	2	0
MG3MDCMFT200	Traditional sports in Kerala	MDC	3	3	3	0	0
MG3SDCMFT203	On-the- Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Semester 4

Course Code	Title of the Course	Type of the Course	Credit	Hours /week	Hour Distribution /week		
					L	P	O
MG4SDCMFT200	Diet and Nutrition in Fitness Programming	SDC	4	4	4	0	0
MG4SDCMFT201	Martial Arts for Fitness & Self Defense - II - Judo & Kurash, Karate	SDC	4	5	3	2	0
MG4SDCMFT202	Principles of injury assessment and Safety planning	SDC	4	5	3	2	0
MG4MPCMFT200	Yoga and Life Style Management	MPC	4	5	3	2	0
MG4SECMFT200	Principles of Fitness Management	SEC	3	3	3	0	0
MG4VACMFT200	Soft Skills for Fitness Professionals	VAC	3	3	3	0	0
MG4INTMFT200	Internship	INT	2	60 HOURS			

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Semester 5

Course Code	Title of the Course	Type of the Course	Credit	Hours /week	Hour Distribution /week		
					L	P	O
MG5SDCMFT300	Fitness facility & Client Management	SDC	4	4	4	0	0
MG5SDEMFT300	Group Fitness & Mind-Body Practices	SDE (Elective papers-selection one)	4	5	3	2	0
MG5SDEMFT301	Massage Manipulation: Theory and Practice						
MG5SDCMFT301	Foundation of Core Strengthening & Pilates	SDC	4	5	3	2	0
MG5SECMFT300	Mastering Sports Skill and Techniques	SEC	3	4	2	2	0
MG5MPCMFT300	Scientific Principles of Fitness	MPC	4	4	4	0	0
MG5VACMFT300	Exercise Science for Women and Children	VAC	3	3	3	0	0

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Semester 6

Course Code	Title of the Course	Type of the Course	Credit	Hours /week	Hour Distribution /week		
					L	P	O
MG6SDEMFT300	Mental Health and Physical Fitness Dynamics	SDE(Elective papers-selection one)	4	5	3	2	0
MG6SDEMFT301	Adventure sports and fitness						
MG6SDCMFT300	Advanced Training designs & Periodization	SDC	4	5	3	2	0
MG6SECMFT300	Officiating and Refereeing in Combat Sports (Judo/ Taekwondo/ Karate)	SEC	3	3	3	0	0
MG6MPCMFT300	Fitness Centre Management	MPC	4	4	4	0	0
MG6VACMFT300	Basics of Geriatric Fitness	VAC	3	3	3	0	0
MG6PRJMFT300	Project	PRJ	4	8	0	8	0

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

SemesterS 7&8

B.VOC HONOURS							
COURSE CODE	TITLE OF THE COURSE	TYPE OF THE COURSE	CREDIT	NO.OF DAYS	CREDIT DISTRIBUTION		
					L	P	O
MG7APPMFT400	APPRENTICESHIP	APP	28	280	0	28	0
	Online	MPC	4				
	Online	MPC	4				
	Online	MPC	4				

BVOC HONOURS WITH RESEARCH							
COURSE CODE	TITLE OF THE COURSE	TYPE OF THE COURSE	CREDIT	NO.OF DAYS	CREDIT DISTRIBUTION		
					L	P	O
MG7RINMFT400	RESEARCH INTERNSHIP	RIN	20	200	0	20	0
	Online	SDC	4				
	Online	SDC	4				
	Online	MPC	4				
	Online	MPC	4				
	Online	MPC	4				

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Job Roles and Qualification Packs for Certificate, Diploma, Bachelor's, and Honours Degrees.

JOB ROLES	NHERQF LEVEL	QPs ALIGNED	SECTOR SKILL
Personal Trainer	5	National Occupational Standards SPF/Q1109	Sports, Physical Education, Fitness & Leisure Skills Council
Group Fitness Trainer	5	National Occupational Standards SPF/Q1110	Sports, Physical Education, Fitness & Leisure Skills Council
Un armed Self Defense Trainer	5	National Occupational Standards SPF/Q1105	Sports, Physical Education, Fitness & Leisure Skills Council
Strength & Conditioning Coach	6	National Occupational Standards SPF/Q1111	Sports, Physical Education, Fitness & Leisure Skills Council
Fitness Center Head	7	National Occupational Standards SPF/Q1108	Sports, Physical Education, Fitness & Leisure Skills Council

SEMESTER I

Semester: 1

Course Code	Title of the Course	Type of the Course	Credit	Hours /week	Hour Distribution /week		
					L	P	O
MG1SDCMFT100	Foundation of Resistance & Functional Training	SDC	4	5	3	2	0
MG1SDCMFT101	Principles of Health & Fitness	SDC	4	4	4	0	0
MG1SDCMFT102	Basics of Practical Yoga and Healing	SDC	4	5	3	2	0
MG1MDCMFT100	First aid and Emergency Care	MDC	3	3	3	0	0
MG1SDCMFT103	On-the Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Foundation of Resistance and Functional Training**
Type of Course	**SDC**
Course Code	**MG1SDCMFT100**
Course Level	100
Course Summary	The Foundations of Resistance and Functional Training course equips students with the basic knowledge and practical skills in strength-based and functional movement training methods. The course covers scientific principles of resistance training, muscle physiology, movement patterns, safety, technique, and program design for health and performance. Through both theoretical and hands-on sessions, students will learn how to apply foundational strength training techniques and functional movement systems to support sport-specific and general fitness goals. This course lays the groundwork for advanced training modules and professional fitness instruction.

Semester	1	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	Basic knowledge of human anatomy and physiology.				

COURSE OUTCOMES (CO)

CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	**Understand the scientific principles underlying resistance and functional training.**	**U**	1
CO2	Identify and demonstrate fundamental resistance training exercises and movement patterns.	**S**	**1**
CO3	Apply the principles of progressive overload, specificity, and periodization in training.	**A**	**2**
CO4	Analyze and correct basic movement and lifting techniques to prevent injury.	**An**	**2,5**
CO5	Design and deliver beginner-level resistance and functional training programs for diverse populations.	**C**	**4,5**
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	1	0	0	0	0	**0**	**0**	**0**	**0**	**0**
CO2	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
CO3	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
CO4	**0**	2	**0**	**0**	2	**0**	**0**	**0**	**0**	**0**
CO5	**0**	0	**0**	**2**	2	**0**	**0**	**0**	**0**	**0**
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT

Module	Units	Course description	Hrs	CO No.
1	1	**Introduction to Resistance and Functional Training**		
	1.1	Definitions: Resistance Training, Functional Training, Strength Training	4	1
	1.2	History and Evolution of Strength Training & Functional Fitness	4	2
	1.3	Types of Muscle Contractions: Isometric, Concentric, Eccentric	4	2
	1.4	Principles of Resistance Training: Overload, Progression, Specificity	5	1
	1.5	Health and Performance Benefits of Strength Training	5	5
2	2	**Musculoskeletal Anatomy and Movement Mechanics**		
	2.1	Major Muscle Groups Involved in Resistance Training	4	1
	2.2	Skeletal System and Joint Movements	4	3
	2.3	Planes of Motion and Basic Biomechanics	5	2
	2.4	Functional Movement Patterns: Push, Pull, Squat, Hinge, Rotate, Gait	5	2
	2.5	Common Movement Dysfunctions and Screening (e.g., FMS)	4	4
3	3	**Resistance Training Techniques and Equipment**		

	3.1	Bodyweight, Free Weights, Machines, Resistance Bands	4	1
	3.2	Correct Lifting Techniques: Deadlift, Squat, Press, Row, Lunge	4	1
	3.3	Spotting and Safety Procedures in the Gym	3	1
	3.4	Warm-up, Cool-down and Mobility Techniques	4	2
	3.5	Modifying Exercises for Beginners and Special Populations	4	3
4	4	**Functional Training Concepts and Program Design**		
	4.1	Defining Functional Training: Core, Stability, Balance	2	3
	4.2	Use of Equipment: Kettlebells, Medicine Balls, Suspension Trainers	2	1
	4.3	Training for Daily Movement Efficiency and Injury Prevention	2	2
	4.4	Periodization Models for Functional Training	3	2
	4.5	Designing Circuit Training and Functional Workouts	3	5
5		**TEACHER SPECIFIC CONTENT**		

<table>
<tr><td rowspan="3">Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
● Lesson plans
● Hands on training</td></tr>
<tr><td>Mode of Assessment</td></tr>
<tr><td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
● Lesson plans
● Hands on training</td></tr>
<tr><td rowspan="2">Mode of Assessment</td><td>Continues Comprehensive Assessment (CCA)
Total Mark - 30

<table><tr><th>Assessment Method (CCA)</th><th>Marks</th></tr><tr><td>Assignment</td><td>15</td></tr><tr><td>Individual Involvement & Viva</td><td>15</td></tr><tr><td>Total</td><td>30</td></tr></table></td></tr>
<tr><td>End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 70
Duration – 3 Hrs

<table><tr><th>Assessment Method (ESE)</th><th>Marks</th></tr><tr><td>Lesson Plan</td><td>30</td></tr><tr><td>Demonstration & Presentation</td><td>30</td></tr><tr><td>Viva</td><td>10</td></tr><tr><td>Total</td><td>70</td></tr></table></td></tr>
</table>

References

1. Baechle, T. R., & Earle, R. W. (Eds.). (2022). NSCA's essentials of strength training and conditioning (4th ed., pp. 245–270). Human Kinetics.
2. Schoenfeld, B. J. (2021). Science and development of muscle hypertrophy (2nd ed., pp. 110–135). Human Kinetics.
3. Boyle, M. (2020). New functional training for sports (2nd ed., pp. 95–115). Human Kinetics.
4. Gentil, P. (2022). Resistance training for health and rehabilitation (pp. 142–160). CRC Press.

Suggested Readings

1. Haff, G. G., & Triplett, N. T. (2022). Essentials of personal training (3rd ed., pp. 310–335). Human Kinetics.
2. Contreras, B., & Schoenfeld, B. J. (2021). The science of hip thrusts and glute building (pp. 88–105). Victory Belt Publishing.
3. Simao, R., & Willardson, J. M. (2021). Advanced resistance training: Exercise prescription and technique (pp. 195–210). Nova Science Publishers.
4. McGill, S. (2020). Back mechanic: The step-by-step McGill method to fix back pain (3rd ed., pp. 150–165).

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc .(Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Principles of Health & Fitness**
Type of Course	**SDC**
Course Code	**MG1SDCMFT101**
Course Level	**100**
Course Summary	The Principles of Health & Fitness course introduces students to the foundational concepts and practical knowledge essential to understanding human health and physical fitness. This course emphasizes a holistic approach, exploring physical, mental, emotional, social, and nutritional aspects of wellness. Students will learn about the components of fitness, the principles of exercise, lifestyle-related diseases, stress management, and the significance of proper nutrition. Through this course, learners will be empowered to develop healthy habits, design basic fitness programs, and promote active lifestyles, preparing them for further specialization in multi-sports and fitness domains.

Semester	1	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		4	0	0	60
Pre-requisites, if any	Learners should have some kind of interest in fitness and health.				

COURSE OUTCOMES (CO)

CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	**Understand and explain the key principles of health and physical fitness**	**U**	1,2
CO2	**Identify and describe the major components of health-related and skill-related fitness.**	**U**	**1,2,3,10**
CO3	**Apply fitness principles to improve personal health and promote active lifestyles.**	**A**	**2,3,4,5,6,7,10**
CO4	**Analyze health risk factors and lifestyle influences on well-being.**	**An**	**1,2,4,5,6,8,10**

CO5	**Design basic health and fitness programs incorporating wellness strategies.**	**C**	**1,2,3,5,6,7,10**
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) Appreciation (Ap			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	0	0	0	0	0	0	0	0
CO2	1	3	2	0	0	0	0	0	0	3
CO3	0	2	3	2	2	3	2	0	0	2
CO4	1	2	0-	1	1	3	0	2	0	3
CO5	2	1	3	0	3	3	1	0	0	2
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Introduction to Health and Fitness**		
	1.1	Definition of Health – WHO Perspective	3	1
	1.2	Dimensions of Health: Physical, Mental, Emotional, Social, and Spiritual	3	1
	1.3	Fitness vs. Health: Understanding the Difference	3	2
	1.4	Benefits of Physical Activity and Fitness on Health	4	1
	1.5	Determinants of Health and Healthy Lifestyle Concepts	4	1
2	2	**Components of Physical Fitness**		
	2.1	Health-Related Components: Cardiorespiratory Endurance, Muscular Strength, Endurance, Flexibility, Body Composition	3	2

	2.2	Skill-Related Components: Agility, Balance, Coordination, Speed, Reaction Time, Power	3	2
	2.3	Principles of Exercise (FITT, Overload, Specificity, Progression)	4	1
	2.4	Common Fitness Assessment Tools	4	2
	2.5	Interpreting Results and Goal Setting	3	2
3	3	**Nutrition and Health**		
	3.1	Basic Nutrients – Carbohydrates, Proteins, Fats, Vitamins, Minerals, and Water	3	1
	3.2	Nutrition for Active Individuals and Athletes	3	2
	3.3	Hydration and Electrolyte Balance	2	1
	3.4	Balanced Diet and Meal Planning	3	2
	3.5	Food Labels, Supplements, and Safe Practices	3	2
4	4	**Mental Health, Stress, and Lifestyle**		
	4.1	Introduction to Mental Health and Emotional Well-being	2	1
	4.2	Stress: Causes, Effects, and Management Techniques	2	2
	4.3	Sleep, Recovery, and Their Role in Fitness	2	2
	4.4	Sedentary Lifestyle Risks and Physical Inactivity	3	1
	4.5	Substance Abuse and Addictions – Fitness Implications	3	1
5		**TEACHER SPECIFIC CONTENT**		

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Lecture (Power Point presentation) ● Demonstration ● Peer teaching ● Lesson plans ● Hands on training	
Assessment Types	**Mode of Assessment**	
	MODE OF ASSESSMENT **Continuous Comprehensive Assessment (CCA) for Theory:30 Marks** CCA -15 marks (Assignment, viva) CCA -15 mark, (Presentation, individual involvement)	
	End Semester Examination (ESE) – **ESE Theory –70 marks (Written examination theory) –** **Duration of Examination 2 hrs –** **Different part of examinations** – **Part 1**, MCQ 10x1=10 (number of questions 12-number of questions to be answered 10) **Part 2** - Short Answer – marks - 5x2=10 (number of questions 7 -number of questions to be answered 5), **Part 3 -** Short Essay - 4x5=20 (number of questions 6-number of questions to be answered 4) **Part 4** - Essay-2 x 15=30 - (number of questions 3-number of questions to be answered 2).	

References

1. Howley, E. T., & Thompson, D. L. (2022). Fitness professional's handbook (8th ed., pp. 150–175). Human Kinetics.
2. Kenney, W. L., Wilmore, J. H., & Costill, D. L. (2023). Physiology of sport and exercise (8th ed., pp. 271–295). Human Kinetics.
3. Fahey, T. D. (2022). Fit & well: Core concepts and labs in physical fitness and wellness (14th ed., pp. 312–330). McGraw-Hill.
4. American College of Sports Medicine. (2022). ACSM's guidelines for exercise testing and prescription (11th ed., pp. 180–200). Wolters Kluwer.

Suggested Reading

1. Gordon, B., & Byrne, H. (2021). Foundations of exercise science and fitness (pp. 225–245). Jones & Bartlett Learning.
2. Campbell, N., & Jones, A. (2021). Mental health and wellbeing in sport and exercise (pp. 110–125). Routledge.
3. Ekkekakis, P. (2023). The measurement of affect, mood, and emotion in exercise psychology (pp. 220–235). Cambridge University Press.
4. Thompson, J. (Ed.). (2023). Worldwide survey of fitness trends 2023. ACSM's Health & Fitness Journal, 27(1), 19–30. https://doi.org/10.1249/FIT.0000000000000805
5. Turner, M. J., & Barker, J. (2021). Sport and exercise psychology: Practitioner case studies (pp. 145–160). Routledge.
6. Bouchard, C., Blair, S. N., & Katzmarzyk, P. T. (2023). Physical activity and health (2nd ed., pp. 190–210). Human Kinetics.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Basics of Practical Yoga and Healing**
Type of Course	**SDC**
Course Code	**MG1SDCMFT102**
Course Level	100
Course Summary	The course "Basics of Practical Yoga and Healing" introduces students to the core principles, philosophy, and practice of yoga and its therapeutic benefits. Students will learn the fundamentals of yogic practices such as asanas (postures), pranayama (breathing techniques), and dhyana (meditation) with a special focus on healing the body and mind. The course blends theory with practical sessions to enable students to experience the transformative and rehabilitative aspects of yoga. It also explores how yoga can complement sports training, injury recovery, stress relief, and overall physical and mental well-being.

Semester	1	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75

Pre-requisites, if any	Students should have basic physical fitness, an interest in holistic health, and a willingness to participate in regular yoga practice. A foundational understanding of the human body and the ability to follow instructions are also recommended.

COURSE OUTCOMES (CO)

CON o	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	**Describe the origin, philosophy, and basic principles of yoga.**	**U**	1,10
CO2	Demonstrate fundamental yogic postures and breathing techniques safely.	**U**	**1**

CO3	Apply yoga as a tool for physical fitness, healing, and mental well-being.	**K**	**10**
CO4	Integrate yoga into multi-sport training routines for recovery and injury prevention.	**C**	**2,3,4**
CO5	Design beginner-level yoga and healing routines for diverse populations.	**C**	**1,10**
:Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**
CO 1	2	0	0	0	0	0	0	0	0	3
CO2	1	0	0	0	0	0	0	0	0	0
CO3	0	0	0	0	0	0	0	0	0	3
CO4	0	2	1	1	0	0	0	0	0	0
CO5	3	0	0	0	0	0	0	0	0	1

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Introduction to Yoga Philosophy and Concept of Healing**		
	1.1	Origin and History of Yoga – Vedic, Pre-classical, and Classical Yoga	4	1
	1.2	Eight Limbs of Yoga (Ashtanga Yoga) – Patanjali Yoga Sutras	4	2
	1.3	Health and Healing in Yogic Perspective – Concept of Pancha Kosha	4	5
	1.4	Role of Yoga in Modern Life – Preventive, Curative and Promotive Health	5	3
	1.5	Difference between Yoga, Exercise, and Physiotherapy	5	4
2	2	**Yogic Physiology and Foundations of Practice**		
	2.1	Yogic Understanding of the Body – Nadis, Chakras, Prana, and Koshas	4	5
	2.2	Introduction to Hatha Yoga and Ashtanga Yoga	4	2
	2.3	Guidelines for Yoga Practice – Contraindications and Precautions	5	1

	2.4	Importance of Breath Awareness and Alignment	5	2
	2.5	Yoga and the Nervous System – Parasympathetic Response and Healing	4	3
3	3	**Asanas for Health, Flexibility, and Recovery**		
	3.1	Standing Asanas – Tadasana, Trikonasana, Veerabhadrasana	4	2
	3.2	Sitting Asanas – Dandasana, Ardha Matsyendrasana, Baddha Konasana	4	2
	3.3	Supine and Prone Asanas – Bhujangasana, Setu Bandhasana, Pavanamuktasana	3	2
	3.4	Balancing and Inversion Asanas – Vrikshasana, Sarvangasana (basic variations)	4	2
	3.5	Use of Props and Modifications for Injuries and Limitations	4	4
4	4	**Breathing, Relaxation, and Meditative Practices**		
	4.1	Pranayama Basics – Nadi Shodhana, Bhramari, Sheetali, Kapalabhati	2	2
	4.2	Relaxation Techniques – Yoga Nidra, Body Scan, Guided Relaxation	2	3
	4.3	Dharana and Dhyana – Techniques for Concentration and Inner Awareness	2	3
	4.4	Mantra Chanting, OM Meditation, and Sound Healing Introduction	3	5
	4.5	Yoga for Emotional Balance and Stress Relief	3	3
5		**TEACHER SPECIFIC CONTENT**		

<table>
<tr><td rowspan="3">Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
● Lesson plans
● Hands on training</td></tr>
<tr><td>Mode of Assessment</td></tr>
<tr><td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
● Lesson plans
● Hands on training</td></tr>
<tr><td rowspan="2">Mode of Assessment</td><td>Continues Comprehensive Assessment (CCA)
Total Mark - 30
<table><tr><th>Assessment Method (CCA)</th><th>Marks</th></tr><tr><td>Assignment</td><td>15</td></tr><tr><td>Individual Involvement & Viva</td><td>15</td></tr><tr><td>Total</td><td>30</td></tr></table></td></tr>
<tr><td>End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 70
Duration – 3 Hrs
<table><tr><th>Assessment Method (ESE)</th><th>Marks</th></tr><tr><td>Lesson Plan</td><td>30</td></tr><tr><td>Demonstration & Presentation</td><td>30</td></tr><tr><td>Viva</td><td>10</td></tr><tr><td>Total</td><td>70</td></tr></table></td></tr>
</table>

References

1. Bhavanani, A. B. (2021). Perspectives on yoga therapy (pp. 85–105). Dhivyananda Creations.
2. Brown, D., & Gerbarg, P. L. (2020). Breathing for health: Science and practice of therapeutic breathing (pp. 120–140). Norton.
3. Swanson, A. (2023). Science of yoga: Understand the anatomy and physiology to perfect your practice (2nd ed., pp. 90–110). DK Publishing.
4. Saraswati, S. S. (2020). Asana pranayama mudra bandha (Revised ed., pp. 275–300). Bihar School of Yoga.

Suggested Readings

1. McCall, T. (2021). Yoga as medicine: The yogic prescription for health and healing (2nd ed., pp. 230–255). Bantam.
2. Lasater, J. H., & Case, M. E. (2021). Restore and rebalance: Yoga for deep relaxation (pp. 115–135). Shambhala.
3. Fishman, L. M., & Saltonstall, C. (2022). Yoga for back pain (pp. 145–165). W. W. Norton & Company.
4. Telles, S., & Naveen, K. V. (2021). Yoga for mental health and well-being (pp. 95–115). Springer. https://doi.org/10.1007/978-981-33-6047-4

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**First Aid and Emergency Care**
Type of Course	**MDC**
Course Code	**MG1MDCMFT100**
Course Level	100
Course Summary	This course is designed to provide undergraduate students from diverse vocational backgrounds with foundational knowledge and practical understanding of First Aid and Emergency Care. Students will learn to assess, respond to, and manage common injuries, medical emergencies, and crisis situations effectively until professional help arrives. The course emphasizes hands-on skills, legal responsibilities, safety, and real-world applications relevant across work environments.

<table>
<tr><td>Semester</td><td>1</td><td colspan="2">Credits</td><td>3</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>45</td></tr>
<tr><td>Pre-requisites, if any</td><td colspan="5">Basic understanding of human body systems (recommended, not mandatory), Willingness to engage in practical simulations and role-plays, Students from all streams (health, sports, education, hospitality, engineering, etc.) are welcome</td></tr>
</table>

COURSE OUTCOMES (CO)

CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	**Identify the principles and responsibilities involved in providing first aid and emergency care.**	K	2,10
CO2	Assess emergency situations and respond appropriately using standard first aid techniques.	K	3,4
CO3	Demonstrate proper procedures for CPR, bleeding control, fracture immobilization, and other life-saving interventions.	An	4,5
CO4	Recognize signs and symptoms of common medical emergencies and initiate immediate care.	U	2,5
CO5	Understand the legal and ethical aspects of providing emergency care in various environments.	U	3,6
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	0	3	0	0	0	0	0	0	0	2
CO2	0	0	2	1	0	0	0	0	0	0
CO3	0	0	0	2	2	0	0	0	0	0
CO4	0	3	0	0	1	0	0	0	0	0
CO5	0	0	2	0	0	3	0	0	0	0

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Module	Units	Course description	Hrs	CO No.
1	1	**Introduction to First Aid and Emergency Care**		
	1.1	Definition, Aims and Importance of First Aid	5	2
	1.2	Roles and Responsibilities of a First Aider	4	1
	1.3	Basic Principles of Emergency Care (DRSABCD)	4	2
	1.4	First Aid Kit: Types, Contents, and Maintenance	4	2
2	2	**Managing Injuries and Bleeding**		
	2.1	Types of Wounds and Bleeding	4	1
	2.2	Control of External and Internal Bleeding	4	2
	2.3	Burns and Scalds – Classification and Management	4	3
	2.4	Bandaging Techniques and Immobilization	5	2
3	3	**Emergency Response to Medical Conditions**		
	3.1	Cardiopulmonary Resuscitation (CPR) and AED	4	1
	3.2	Choking, Drowning, and Asphyxia – First Aid Measures	3	2
	3.3	Seizures, Fainting, Diabetic Emergencies	2	2
	3.4	Heat Stroke, Hypothermia, and Allergic Reactions	2	2
4		**TEACHER SPECIFIC CONTENT**		

Teaching and Learning Approach

Classroom Procedure (Mode of transaction)

- Lecture (Power Point presentation)
- Demonstration
- Peer teaching
- Lesson plans
- Hands on training

Mode of Assessment

Continues Comprehensive Assessment (CCA)

Total Mark - 25

Assessment Method (CCA)	Marks
Assignment	10
Individual Involvement & Viva	15
Total	25

Assessment Type

End Semester Examination (ESE)

Assessment Method – Practical Examination

Total Marks – 50

Duration – 2 Hrs

Assessment Method (ESE)	Marks
Lesson Plan	20
Demonstration & Presentation	20
Viva	10
Total	50

References

1. American Red Cross. (2021). First aid/CPR/AED participant's manual (7th ed., pp. 75–95). StayWell.
2. Eberst, R. (2020). The essentials of first aid and CPR (pp. 60–80). Springer Publishing.
3. Ellis, H. (2023). Anatomy for first aid and CPR providers (pp. 55–70). Elsevier.
4. National Safety Council. (2022). First aid and CPR handbook (5th ed., pp. 100–120). McGraw Hill.
5. Nims, B. (2023). Emergency medical response workbook (pp. 130–150). Jones & Bartlett Learning.

Suggested Readings

1. Porth, C. M., & Matfin, G. (2021). Pathophysiology: Concepts of altered health states (10th ed., pp. 1120–1140). Lippincott Williams & Wilkins.
2. Thygerson, A. L., & Thygerson, S. M. (2021). First aid, CPR, and AED essentials (pp. 140–160). Jones & Bartlett Learning.
3. Wilkinson, J. M., & Treas, L. S. (2022). Fundamentals of nursing: Volume 2 – Emergency care (pp. 205–225). F.A. Davis Company.

SEMESTER II

Semester: 2

Course Code	Title of the Course	Type of the Course	Credit	Hours /week	Hour Distribution /week		
					L	P	O
MG2SDCMFT100	Human Body: Structure, function & Energy Metabolism	SDC	4	4	4	0	0
MG2SDCMFT101	Fitness Recovery: Massage, Wellness, and Recreation	SDC	4	5	3	2	0
MG2SDCMFT102	Martial Arts for Fitness & Self-defense - I - Taekwondo, Krav Maga	SDC	4	5	3	2	0
MG2MDCMFT100	Self Defense for Fitness	MDC	3	4	2	2	0
MG2SDCMFT103	On-the-Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Human Body: Structure, Function & Energy Metabolism**
Type of Course	**SDC**
Course Code	**MG2SDCMFT100**
Course Level	100
Course Summary	This course provides an in-depth understanding of human anatomy, physiology, kinesiology, biomechanics, and energy metabolism, highlighting how these biological systems support physical movement and fitness. It focuses on the structure and function of body systems, principles of movement, and the biochemical processes that fuel muscular work. Designed for students in fitness and sports training, this course bridges the gap between biological science and practical application in sports, exercise, and health. Emphasis is placed on applying knowledge to injury prevention, strength development, and performance enhancement.

Semester	2	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		4	0	0	60
Pre-requisites, if any	Basic knowledge of physical fitness and health education				

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	**Describe the structure and function of major systems of the human body.**	**U**	1,2,5
CO2	Explain the physiological processes involved in movement, exercise, and recovery.	**U**	**3,5,10**
CO3	Apply biomechanical and kinesiological principles to sports and fitness practices.	**A**	**3,5**
CO4	Understand energy systems and metabolic responses to different types of exercise.	**U**	**3,4,5**
CO5	Integrate anatomical, physiological, and biomechanical knowledge to optimize physical performance and prevent injury.	**A**	**3,4,5**
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	1	1	0	0	1	0	0	0	0	0
CO2	**0**	0	1	0	1	0	0	0	0	3
CO3	0	**0**	2	0	**1**	0	0	0	0	0
CO4	0	0	**1**	1	2	0	0	0	0	0
CO5	0	0	1	3	2	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Human Anatomy – Structure and Systems**		
	1.1	Introduction to Anatomy: Planes, Positions, Terms	3	1
	1.2	Skeletal System: Types of Bones, Axial and Appendicular Skeleton	3	3
	1.3	Muscular System: Types of Muscles, Major Muscle Groups	3	1
	1.4	Joints and Ligaments: Classification, Joint Movements	4	2
	1.5	Nervous System and Reflexes Related to Movement	4	1
2	2	**Human Physiology in Fitness Context**		
	2.1	Overview of Cell Physiology and Homeostasis	3	2
	2.2	Cardiovascular System: Heart Function, Circulation in Exercise	3	1
	2.3	Respiratory System: Lung Volumes, VO_2 Max, Breathing Mechanics	4	1
	2.4	Endocrine System: Hormones and Exercise Response	4	1
	2.5	Thermoregulation, Hydration and Electrolyte Balance in Activity	3	2
3	3	**Biomechanics of Movement**		
	3.1	Definition, Scope and Application of Biomechanics in Sports	3	3

	3.2	Force, Motion, Torque and Levers in the Human Body	3	3
	3.3	Center of Gravity, Balance, and Stability	2	3
	3.4	Joint Kinetics and Injury Mechanisms	3	5
	3.5	Biomechanical Analysis of Functional Movements	3	2
4	4	**Kinesiology – Principles of Movement**		
	4.1	Principles of Kinesiology and Human Motion	2	3
	4.2	Muscle Action – Agonist, Antagonist, Stabilizers	2	2
	4.3	Kinematic Chains – Open vs Closed	2	3
	4.4	Posture, Gait, and Movement Efficiency	3	2
	4.5	Kinesiological Considerations for Lifting, Running, and Throwing	3	3
5		**TEACHER SPECIFIC CONTENT**		

Teaching and Learning Approach **Assessment Types**	**Classroom Procedure (Mode of transaction)** ● Lecture (Power Point presentation) ● Demonstration ● Peer teaching ● Lesson plans ● Hands on training
	Mode of Assessment
	MODE OF ASSESSMENT Continuous Comprehensive Assessment (CCA) **CCA for Theory : 30 Marks** CCA -15 Mark(Assignment, viva) CCA -15 mark, (Presentation, individual involvement)

Assessment Type	**End Semester Examination (ESE) –** **ESE Theory –70 marks (Written examination theory) –** **Duration of Examination 2 hrs –** **Different part of examinations** – **Part 1**, MCQ 10x1=10 (number of questions 12-number of questions to be answered 10) **Part 2** - Short Answer – marks - 5x2=10 (number of questions 7 -number of questions to be answered 5) **Part 3 -** Short Essay - 4x5=20 (number of questions 6-number of questions to be answered 4) **Part 4** - Essay-2 x 15=30 - (number of questions 3-number of questions to be answered 2)

References

1. Floyd, R. T. (2023). Manual of structural kinesiology (21st ed., pp. 125–145). McGraw-Hill Education.
2. Hall, S. J. (2021). Basic biomechanics (9th ed., pp. 210–230). McGraw-Hill Education.
3. Hoffman, J. (2020). Physiological aspects of sport training and performance (3rd ed., pp. 180–200). Human Kinetics.
4. McArdle, W. D., Katch, F. I., & Katch, V. L. (2022). Exercise physiology: Nutrition, energy, and human performance (9th ed., pp. 345–370). Wolters Kluwer.

Suggested Readings

1. Neumann, D. A. (2022). Kinesiology of the musculoskeletal system: Foundations for rehabilitation (3rd ed., pp. 300–320). Elsevier.
2. Powers, S. K., & Howley, E. T. (2020). Exercise physiology: Theory and application to fitness and performance (10th ed., pp. 240–260). McGraw-Hill Education.
3. Wilmore, J. H., Costill, D. L., & Kenney, W. L. (2021). Physiology of sport and exercise (7th ed., pp. 220–240). Human Kinetics.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Fitness Recovery: Massage, Wellness, and Recreation**
Type of Course	**SDC**
Course Code	**MG2SDCMFT101**
Course Level	100
Course Summary	This course introduces students to **practical recovery methods** essential for fitness and sports performance. It covers **recreational activities** (indoor and outdoor), **massage techniques**, **relaxation and stress-reduction methods**, and **wellness principles** to support holistic recovery. Emphasis is placed on active participation, skill development, and implementing strategies that promote physical and mental well-being in diverse fitness populations.

Semester	2	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	Students should have **basic awareness of fitness and physical activity** concepts introduced during their foundational course or orientation semester.				

EXPECTED COURSE OUTCOMES (CO)

CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	Demonstrate various indoor and outdoor recreational activities that aid recovery and well-being.	**K**	1
CO2	Apply basic massage techniques for muscle relaxation and post-exercise recovery.	**An**	**2**
CO3	Practice and instruct relaxation methods such as breathing, meditation, and guided stretching.	**U**	**4,5**
CO4	Organize recovery-oriented recreational sessions in different environments.	**U**	**2**
CO5	Incorporate wellness strategies, including hydration, rest, and sleep hygiene, into fitness recovery plans.	**U**	**2,5**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	0	0	0	0	0	0	0	0	0
CO2	0	2	0	0	0	0	0	0	0	0
CO3	0	0	0	3	2	0	0	0	0	0
CO4	0	2	0	0	0	0	0	0	0	0
CO5	0	3	0	0	2	0	0	0	0	0

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Module	Units	Course description	Hrs	CO No.
1	1	**Introduction to Fitness Recovery and Wellness**		
	1.1	Concept and Importance of Recovery in Fitness	4	1
	1.2	Components of Wellness: Physical, Mental, Social	4	1
	1.3	Active vs. Passive Recovery Techniques	4	2
	1.4	Role of Sleep, Hydration, and Nutrition in Recovery	5	2
	1.5	Recovery Tracking Tools and Apps (Basics)	5	2
2	2	**Recreational Activities for Recovery (Indoor & Outdoor)**		
	2.1	Aquatic-Based Recovery Activities (If available: Pool Walking, Hydrotherapy)	4	1
	2.2	Indoor Games (Yoga Ball, Table Tennis, Dance Fitness, Resistance Games)	4	4
	2.3	Introduction to Therapeutic Recreation	5	4
	2.4	Outdoor Activities (Nature Walks, Team Games, Cycling, Frisbee)	5	4
	2.5	Designing Recreation Sessions for Recovery	4	4
3	3	**Massage and Manual Therapy Techniques**		
	3.1	Introduction to Massage and Bodywork	4	1
	3.2	Swedish Massage Basics: Effleurage, Petrissage, Tapotement	4	2

	3.3	Self-Massage Techniques (Foam Roller, Massage Ball)	3	2
	3.4	Contraindications and Safety in Massage	4	4
	3.5	Practice Sessions in Pair or Group Format	4	4
4	4	**Relaxation and Recovery Techniques**		
	4.1	Deep Breathing and Diaphragmatic Breathing	2	2
	4.2	Progressive Muscle Relaxation (PMR)	2	2
	4.3	Guided Meditation and Visualization	2	4
	4.4	Static and Dynamic Stretching for Recovery	3	5
	4.5	Music and Aromatherapy for Relaxation (Demonstration Based)	3	3
5		**TEACHER SPECIFIC CONTENT**		

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** **Mode of Assessment** **Classroom Procedure (Mode of transaction)** ● Lecture (Power Point presentation) ● Demonstration ● Peer teaching ● Lesson plans ● Group Activities
Mode of Assessment	**Continues Comprehensive Assessment (CCA)** Total Mark - 30

Assessment Method (CCA)	**Marks**
Assignment	15
Individual Involvement & Viva	15
Total	30

End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 70
Duration – 3 Hrs

Assessment Method (ESE)	Marks
Lesson Plan	30
Demonstration & Presentation	30
Viva	10
Total	70

References

1. Howley, E. T., & Thompson, D. L. (2022). *Fitness professional's handbook* (8th ed.). Human Kinetics. **(pp. 352–390)**
2. Brawley, L. R. (2023). *Fitness management: A strategic approach*. Routledge. **(pp. 280–305)**
3. Nelson, M. E., & Rejeski, W. J. (2021). *Physical activity and public health in older adults*. ACSM. **(pp. 112–138)**
4. Beck, M. (2021). *Healing massage: An essential guide*. DK Publishing. **(pp. 50–97)**
5. Fahey, T. D., Insel, P. M., & Roth, W. T. (2022). *Fit & well: Core concepts and labs in physical fitness and wellness* (14th ed.). McGraw-Hill. **(pp. 164–192)**

Suggested Readings

1. Clay, J., & Pounds, D. M. (2020). *Basic clinical massage therapy* (3rd ed.). Lippincott Williams & Wilkins. **(pp. 118–145)**
2. Seaward, B. L. (2021). *Managing stress: Principles and strategies for health and well-being* (10th ed.). Jones & Bartlett. **(pp. 78–106)**
3. Greenberg, J. S., Dintiman, G. B., & Oakes, B. M. (2022). *Health & wellness* (13th ed.). Pearson. **(pp. 202–240)**

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc.(Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Martial Arts for Fitness & Self-Defense – I (Taekwondo & Krav Maga)**
Type of Course	**SDC**
Course Code	**MG2SDCMFT102**
Course Level	100
Course Summary	This course introduces students to martial arts as a functional approach to fitness and self-defense, focusing on Taekwondo and Krav Maga. It combines traditional martial arts discipline and modern tactical techniques to improve strength, agility, mental focus, coordination, and safety awareness. Students will explore the philosophy, biomechanics, techniques, and fitness application of both martial arts while gaining hands-on experience through drills, movement patterns, and simulated scenarios. This course develops not only physical conditioning but also confidence, stress control, and resilience—essential for both fitness professionals and everyday life.

Semester	2	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	Students must have completed introductory fitness and movement science courses. Basic mobility, posture control, and aerobic fitness are expected. No prior martial arts experience required.				

COURSE OUTCOMES (CO)

CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	**Describe the history, principles, and objectives of Taekwondo and Krav Maga**	**K**	2,5
CO2	Demonstrate fundamental techniques from Taekwondo and Krav Maga with proper form and control.	**U**	**3,5,8**
CO3	Apply martial arts-based drills to improve cardiovascular fitness, strength, flexibility, and coordination.	**U**	**2,3,5**
CO4	Utilize Krav Maga principles for basic self-defense and situational awareness.	**U**	**3,5,8**
CO5	Design and lead fitness routines incorporating martial arts techniques for individual and group training.	**U**	**2,5,10**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	0	2	0	0	1	0	0	0	0	0
CO2	0	0	2	0	2	0	0	2	0	0
CO3	0	3	1	0	2	0	0	0	0	0
CO4	0	0	2	0	1	0	0	3	0	0
CO5	0	2	0	0	3	0	0	0	0	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Module	Units	Course description	Hrs	CO No.
1	1	**Introduction to Martial Arts for Fitness and Wellness**		
	1.1	Definition and Scope of Martial Arts in Fitness	4	1
	1.2	Comparison of Traditional and Modern Martial Arts Forms	4	3
	1.3	Principles of Mind-Body Training and Discipline	4	4
	1.4	Warm-up, Flexibility & Conditioning for Martial Arts Practice	5	1
	1.5	Benefits of Martial Arts on Mental Health and Emotional Control	5	1
2	2	**Fundamentals of Taekwondo for Fitness**		
	2.1	History and Philosophy of Taekwondo	4	1
	2.2	Taekwondo Stances, Footwork, and Forms (Basic Taegeuk)	4	1
	2.3	Basic Kicks: Front Kick, Side Kick, Roundhouse Kick	5	1
	2.4	Basic Punches, Strikes, and Blocks	5	2
	2.5	Taekwondo Fitness Drills for Strength, Speed, and Endurance	4	4
3	3	**Introduction to Krav Maga and Tactical Fitness**		
	3.1	History and Principles of Krav Maga	4	2
	3.2	Fighting Stance, Movement, and Striking Combinations	4	4

	3.3	Defense against Common Attacks (Pushes, Holds, Grabs)	3	5
	3.4	Tactical Conditioning and Stress Drills	4	2
	3.5	Krav Maga for Real-World Scenarios: Awareness and De-escalation	4	5
4	4	**Martial Arts Conditioning and Cross-training**		
	4.1	Cardiovascular Conditioning with Martial Arts	2	5
	4.2	Strength Training for Kick and Strike Power	2	5
	4.3	Core Training and Balance in Martial Arts	2	5
	4.4	Plyometric and Mobility Drills for Agility and Reaction	3	5
	4.5	Recovery Techniques and Injury Prevention in Combat Fitness	3	5
5		**TEACHER SPECIFIC CONTENT**		

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Lecture (Power Point presentation) ● Demonstration ● Peer teaching ● Lesson plans ● Hands on training
	Mode of Assessment
Mode of Assessment	**Continues Comprehensive Assessment (CCA)** Total Mark - 30

Assessment Method (CCA)	Marks
Assignment	15
Individual Involvement & Viva	15
Total	30

End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 70
Duration – 3 Hrs

Assessment Method (ESE)	**Marks**
Lesson Plan	30
Demonstration & Presentation	30
Viva	10
Total	70

References

1. Cho, Y. H., & Park, J. S. (2021). Taekwondo: The state of the art (pp. 115–135). Human Kinetics.
2. Dreiling, D. (2021). Martial arts and wellness: Training the body, mind, and spirit (pp. 90–110). Routledge.
3. Garcia, D., & Franks, J. (2022). Krav Maga for beginners: A step-by-step guide (pp. 60–80). Ulysses Press.
4. Itay, G., & Lichtenfeld, E. (2022). Krav Maga: Tactical survival for self-defense (pp. 135–150). Black Belt Publishing.
5. Miller, D. L. (2023). Fitness through martial arts: Functional conditioning for real life (pp. 100–120). Human Kinetics.
6. Mindlin, G. (2020). Psychology and philosophy of martial arts (pp. 70–90). Springer.
7. Seo, K. H. (2020). Complete Taekwondo poomsae: The official forms (pp. 55–75). Turtle Press.
8. Singh, R., & Rathi, A. (2022). Applied martial arts for physical education and fitness (pp. 80–95). Khel Sahitya Kendra.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Self-defense for Fitness**
Type of Course	**MDC**
Course Code	**MG2MDCMFT100**
Course Level	100
Course Summary	This course is designed to equip students with foundational knowledge and skills in self-defense techniques that enhance both physical fitness and mental resilience. Emphasis is placed on situational awareness, body mechanics, escape strategies, and fitness-based combat routines suitable for different threat scenarios. By the end of the course, students will gain the confidence to defend themselves while maintaining a high level of physical conditioning.

Semester	2	**Credits**		3	**Total Hours**
Course details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		2	1	0	**60**
Pre-requisites, if any	Basic level of physical fitness, willingness to engage in physical activity, and interest in personal safety and mental preparedness. No prior martial arts experience is required.				

COURSE OUTCOMES (CO)

CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	**Identify and assess real-life self-defense situations and respond appropriately.**	An	1
CO2	Demonstrate basic to intermediate-level physical self-defense techniques.	A	2
CO3	Improve overall fitness, strength, and flexibility through functional self-defense drills	A	2
CO4	Developmental preparedness and emotional control during threatening scenarios.	An	2
CO5	Integrate self-defense strategies into a regular fitness routine for long-term personal safety.	A	3,6,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX										
CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	0	0	0	0	0	0	0	0	0
CO2	1	2	0	0	0	0	0	0	0	0
CO3	0	2	0	0	0	0	0	0	0	0
CO4	0	2	0	0	0	0	0	0	0	0
CO5	0	0	2	0	0	2	0	0	0	1
	‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).									

COURSE CONTENT				
Module	**Units**	**COURSE DESCRIPTION**	**Hrs**	**CO No.**
1	1	**Introduction to Self-Defense and Fitness Integration**		
	1.1	Understanding Self-Defense: Concepts, Myths & Realities	5	1
	1.2	Legal and Ethical Considerations in Self-Defense	5	1
	1.3	Role of Fitness in Personal Safety	5	2
	1.4	Principles of Body Awareness and Movement	5	1
2	2	**Physical Conditioning for Self-Defense**		
	2.1	Functional Fitness for Defensive Readiness	5	1
	2.2	Warm-Up, Mobility, and Flexibility Drills	5	2
	2.3	Core Strengthening & Endurance Workouts	5	2
	2.4	Speed, Agility & Reflex Development	5	3
3	3	**Fundamental Self-Defense Techniques**		
	3.1	Defensive Stance, Guarding & Movement	5	2
	3.2	Escape Techniques: Grabs, Holds, and Chokes	5	3
	3.3	Blocking, Evasion, and Redirection Tactics	5	2
	3.4	Basic Striking Skills: Palm Heel, Elbow, Knee, and Foot	5	2
4		**TEACHER SPECIFIC CONTENT**		

<table>
<tr><td>Teaching and Learning Approach</td><td></td><td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
● Lesson plans
● Hands on training</td></tr>
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Total Mark - 25
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Assessment Method – Practical Examination
Total Marks – 50
Duration – 2 Hrs
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</table>

References

1. Blauer, T. (2020). Know fear: The psychology of self-defense (pp. 85–105). Blauer Tactical Systems.
2. Gracie, R., & Danaher, J. (2022). Basic Jiu-Jitsu for self defense (pp. 140–160). Victory Belt Publishing.
3. Higginbotham, A. (2022). Mental conditioning for self-defense: Staying calm and focused in confrontation (pp. 100–120). Reality Press.
4. Johnson, L. (2023). Functional training for self defense and martial arts (pp. 125–145). Human Kinetics.
5. Kelly, C. (2020). Smart self-defense: A practical personal safety guide for women and men (pp. 60–75). Independence Press.

Suggested Readings

6. Lee, A. (2023). Fitness and fighting: The hybrid approach to personal defense (pp. 90–110). Tactical Fit Publications.
7. MacYoung, M. (2021). In the name of self-defense: What it costs, when it's worth it (pp. 210–230). Conflict Communications.
8. Urban, S. (2021). The self-defense handbook: The best street fighting moves and self-defense techniques (pp. 115–135). Urban Survival.

SEMESTER III

Semester 3

Course Code	Title of the Course	Type of the Course	Credit	Hours /week	Hour Distribution /week		
					L	P	O
MG3SDCMFT200	Foundation of Personal Training and Exercise Programming	SDC	4	4	4	0	0
MG3SDCMFT201	Applied Aerobic fitness and Rhythmic workouts	SDC	4	5	3	2	0
MG3SDCMFT202	Fitness Assessment & Evaluation	SDC	4	5	3	2	0
MG3MPCMFT200	Self Defense Training and Fitness	MPC	4	5	3	2	0
MG3MDCMFT200	Traditional sports in Kerala	MDC	3	3	3	0	0
MG3SDCMFT203	On-the- Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Foundation of Personal Training and Exercise Programming**
Type of Course	**SDC**
Course Code	**MG3SDCMFT200**
Course Level	200
Course Summary	This course introduces students to the essential foundations of personal training and exercise programming, helping them understand how to assess, plan, and guide individuals in achieving fitness goals. It covers core principles such as client screening, fitness assessments, individualized goal setting, training methodologies, and the creation of structured, progressive exercise programs. Designed for undergraduate learners pursuing careers in fitness, this course helps students bridge theory with practical application in personal fitness training scenarios, with a focus on communication, safety, and adaptability for diverse populations.

Semester	3	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		4	0	0	60
Pre-requisites, if any	Students must have basic knowledge of human body systems, physical fitness concepts, and movement science (e.g., completion of "Principles of Physical Fitness & Well-being" and "Human Body: Structure to Functioning").				

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	**Explain the scope, ethics, and responsibilities of a personal trainer.**	**K**	1,2,5
CO2	Conduct basic health screening and fitness assessments for individuals.	**U**	**3,5,10**
CO3	Describe the principles of exercise prescription and progression.	**U**	**3,5**
CO4	Develop individualized fitness programs based on client needs and goals.	**U**	**3,4,5**
CO5	Identify different training methods and their application in one-to-one training environments.	**A**	**3,4,5**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	2	0	0	1	0	0	0	0	0
CO2	0	0	3	0	2	0	0	0	0	2
CO3	0	0	2	0	1	0	0	0	0	0
CO4	0	0	2	1	2	0	0	0	0	0
CO5	0	0	1	2	3	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT

Module	Units	Course description	Hrs	CO No.
1	1	**Introduction to Personal Training**		
	1.1	Definition and Scope of Personal Training	3	1
	1.2	Roles, Skills, and Responsibilities of a Personal Trainer	3	1
	1.3	Professional Ethics and Code of Conduct	3	1
	1.4	Communication and Motivation in One-to-One Training	4	1
	1.5	Types of Clients: Age, Fitness Levels, and Special Needs	4	1
2	2	**Health Screening and Client Assessment**		
	2.1	Pre-Exercise Health Screening (PAR-Q, Risk Stratification)	3	1
	2.2	Anthropometric Measurements (BMI, Waist-Hip Ratio, Body Fat %)	3	4
	2.3	Functional Movement Screening (FMS) Basics	4	2
	2.4	Cardiovascular, Muscular, and Flexibility Assessments	4	4
	2.5	Interpreting Results for Exercise Planning	3	2
3	3	**Principles of Exercise Programming**		

	3.1	FITT Principle and Exercise Prescription	3	1
	3.2	Training Load, Progression, and Recovery	3	1
	3.3	Periodization: Micro, Meso, Macro Cycles	2	4
	3.4	Goal Setting: SMART Goals for Fitness Clients	3	2
	3.5	Individualization, Adaptation, and Overload Concepts	3	2
4	4	**Exercise Techniques and Training Methods**		
	4.1	Aerobic and Anaerobic Training Methods	2	2
	4.2	Resistance Training Techniques: Machines, Free Weights, Bands	2	4
	4.3	Flexibility and Mobility Drills	2	5
	4.4	Core Stability and Balance Training	3	4
	4.5	Functional and Corrective Exercise Basics	3	4
5		**TEACHER SPECIFIC CONTENT**		

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Lecture (Power Point presentation) ● Demonstration ● Peer teaching ● Lesson plans ● Hands on training
	Mode of Assessment
	Continues Comprehensive Assessment (CCA) Total Mark - 30

Assessment Method (CCA)	**Marks**
Assignment	15
Individual Involvement & Viva	15
Total	30

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Assessment Method – Practical Examination
Total Marks – 70
Duration – 2 hrs

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References

1. Acevedo, E. O. (2020). ACSM's advanced exercise physiology (2nd ed., pp. 410–430). Wolters Kluwer.
2. American College of Sports Medicine. (2022). ACSM's guidelines for exercise testing and prescription (11th ed., pp. 230–250). Wolters Kluwer.
3. Baechle, T. R., & Earle, R. W. (2022). NSCA's essentials of personal training (3rd ed., pp. 310–330). Human Kinetics.
4. Coburn, J. W., & Malek, M. H. (2020). NSCA's essentials of sport science (pp. 180–200). Human Kinetics.
5. Franks, B. D., & Howley, E. T. (2021). Fitness professional's handbook (8th ed., pp. 275–295). Human Kinetics

Suggested Readings.

1. Gentil, P. (2022). Strength training for all body types: The science of lifting and levers (pp. 125–145). Victory Belt Publishing.
2. Lagally, K. M. (2020). Fitness programming and physical education (pp. 210–230). Cognella Academic Publishing.
3. Schoenfeld, B. J. (2021). Science and development of muscle hypertrophy (2nd ed., pp. 150–170). Human Kinetics.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Applied Aerobic Fitness and Rhythmic Workouts**
Type of Course	**SDC**
Course Code	**MG3SDCMFT201**
Course Level	200
Course Summary	This course provides foundational and practical knowledge of aerobic fitness and rhythmic workout techniques such as aerobics, Zumba, kettlebell workouts, resistance band training, and rope exercises. Students will explore cardiovascular conditioning through structured rhythmic movement and music-based training. The course emphasizes physiological benefits, technique precision, choreography, and safe exercise progression. Through lectures and hands-on practical sessions, learners will gain the competence to perform and instruct rhythmic group workouts in fitness and community settings.

Semester	3	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	Students should have completed introductory courses on physical fitness and movement science, and must possess basic cardiovascular and motor fitness.				

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	**Describe the principles and benefits of aerobic and rhythmic fitness training.**	U	10
CO2	Demonstrate competence in performing aerobic workouts, Zumba, kettlebell, rope, and resistance band exercises.	S	1
CO3	Apply music, rhythm, and tempo appropriately to group fitness routines.	A	1,10

CO4	Develop and lead safe and effective rhythmic workouts for varied populations.	**U**	**1**
CO5	Assess and monitor intensity, progression, and safety during aerobic exercise sessions.	**A**	**1**
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	0	0	0	0	0	0	0	0	0	3
CO2	1	0	0	0	0	0	0	0	0	0
CO3	2	0	0	0	0	0	0	0	0	2
CO4	3	0	0	0	0	0	0	0	0	0
CO5	2	0	0	0	0	0	0	0	0	0

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Foundations of Aerobic Fitness**		
	1.1	Definition and Components of Aerobic Fitness	4	2
	1.2	Energy Systems in Aerobic Activity	4	1
	1.3	Benefits of Cardiorespiratory Endurance Training	4	1
	1.4	Monitoring Intensity (HR Zones, RPE Scale, Talk Test)	5	2
	1.5	Safety Guidelines and Contraindications for Aerobic Exercise	5	2
2	2	**Structured Aerobics and Choreography**		
	2.1	Types of Aerobic Exercise (High/Low Impact, Step, Dance-Based)	4	2
	2.2	Warm-Up and Cool-Down Techniques	4	1
	2.3	Basic to Intermediate Aerobic Choreography Patterns	5	2
	2.4	Music Selection, Beats per Minute (BPM), and Cueing	5	2

	2.5	Aerobic Session Planning and Modification Techniques	4	1
3	3	**Rhythmic Fitness – Zumba and Dance Workouts**		
	3.1	Origins and Principles of Zumba and Dance Fitness	4	1
	3.2	Latin and World Dance Styles in Zumba (Salsa, Reggaeton, Merengue)	4	2
	3.3	Safety, Modifications, and Movement Progressions	3	2
	3.4	Rhythm, Beat, and Expression in Group Workouts	4	2
	3.5	Planning a Zumba/Dance Fitness Class	4	2
4	4	**Resistance-Based Aerobic Tools**		
	4.1	Kettlebell Training for Endurance and Rhythm	2	2
	4.2	Battle Rope Drills: Full-Body Conditioning	2	1
	4.3	Resistance Band Training for Mobility and Strength	2	1
	4.4	Circuit Design Using Aerobic Tools	3	2
	4.5	Functional Applications for Sport and Fitness Clients	3	2
5		**TEACHER SPECIFIC CONTENT**		

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)**
	Mode of Assessment
	Classroom Procedure (Mode of transaction) ● Lecture (Power Point presentation) ● Demonstration ● Peer teaching ● Lesson plans ● Group Activities

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Total Mark - 30

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<tr><td>Individual Involvement & Viva</td><td>15</td></tr>
<tr><td>Total</td><td>30</td></tr>
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<td>End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 70
Duration – 3 hrs

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<tr><td>Lesson Plan</td><td>30</td></tr>
<tr><td>Demonstration & Presentation</td><td>30</td></tr>
<tr><td>Viva</td><td>10</td></tr>
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References

1. Coburn, J. W., & Malek, M. H. (2021). NSCA's essentials of personal training (3rd ed., pp. 300–320). Human Kinetics.
2. Gentil, P. (2022). Strength training for all body types: The science of lifting and levers (pp. 140–160). Victory Belt Publishing.
3. Howley, E. T., & Franks, B. D. (2021). Health fitness instructor's handbook (6th ed., pp. 250–270). Human Kinetics.
4. McGill, S. M. (2020). Back mechanic: The step-by-step guide to help your clients avoid back pain (pp. 165–185). Backfitpro.

Suggested Readings

1. Pivarnik, J. M., & Hill, D. W. (2020). Cardiorespiratory fitness in physical activity and exercise science (pp. 100–120). Routledge.
2. Schoenfeld, B. J. (2021). Science and development of muscle hypertrophy (2nd ed., pp. 175–195). Human Kinetics.
3. Thompson, W. R. (2022). Fitness professional's handbook (8th ed., pp. 220–240). Human Kinetics.
4. Zumba Fitness LLC. (2021). Zumba instructor training manual (pp. 80–95). Zumba Fitness.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc.(Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Fitness Assessment & Evaluation**
Type of Course	**SDC**
Course Code	**MG3SDCMFT202**
Course Level	200
Course Summary	The course "Fitness Assessment & Evaluation" equips students with theoretical knowledge and practical skills required to conduct comprehensive fitness testing and interpret results for various populations. The course emphasizes the use of scientifically valid and reliable testing methods to evaluate components such as cardiovascular fitness, muscular strength, endurance, flexibility, and body composition. Students will learn how to administer tests, interpret data, and use outcomes to design and modify fitness programs effectively.

Semester	3	**Credits**	4		**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	Students must have completed courses such as Principles of Physical Fitness & Well-being and Human Body: Structure to Functioning & Energy Metabolism, and possess basic knowledge of anatomy, physiology, and fitness training principles.				

COURSE OUTCOMES (CO)

CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	**Understand the purpose and scientific basis of physical fitness assessments.**	**U**	1,10
CO2	Perform standardized fitness assessments for all components of health- and skill-related fitness.	**U**	**1**
CO3	Accurately measure, record, and evaluate test results.	**A**	**2,4,5**
CO4	Interpret fitness assessment results and provide recommendations for exercise programming.	**E**	**2,3,4**
CO5	Design and conduct safe and appropriate assessment protocols for special populations.	**C**	**1,10**
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**
CO 1	1	0	0	0	0	0	0	0	0	3
CO2	**2**	**0**	0	**0**	**0**	0	0	0	0	0
CO3	0	1	0	2	2	0	0	0	0	0
CO4	0	2	1	2	0	0	0	0	0	0
CO5	2	0	0	0	0	0	0	0	0	1

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Introduction to Fitness Assessment**		
	1.1	Definition, Objectives, and Scope of Fitness Assessment	4	1
	1.2	Principles of Measurement and Evaluation in Fitness	4	2
	1.3	Validity, Reliability, and Objectivity in Testing	4	5
	1.4	Pre-Test Screening and Risk Stratification (e.g., PAR-Q)	5	3
	1.5	Ethical and Safety Considerations in Fitness Testing	5	4
2	2	**Assessment of Body Composition & Anthropometry**		
	2.1	Anthropometric Measurements: Height, Weight, Circumference	4	2
	2.2	BMI, Waist-Hip Ratio, and WHtR	4	2
	2.3	Skinfold Thickness Measurements and Use of Calipers	5	5
	2.4	Bioelectrical Impedance and Other Body Composition Methods	5	1
	2.5	Interpretation and Application of Body Composition Data	4	3
3	3	**Cardiovascular and Muscular Fitness Testing**		
	3.1	Submaximal and Maximal Cardiorespiratory Tests (e.g., Cooper Test, 3-min Step Test)	4	1
	3.2	Muscular Strength Testing: 1-RM, Handgrip, Isokinetic Testing	4	2
	3.3	Muscular Endurance Tests: Push-Up, Sit-Up, Plank Hold	3	5
	3.4	Heart Rate Monitoring, Blood Pressure Measurement	4	1
	3.5	Interpreting Norms and Risk Levels for Fitness Variables	4	4
4	4	**Flexibility, Mobility, and Posture Assessment**		
	4.1	Importance and Types of Flexibility Assessments	2	1
	4.2	Sit-and-Reach Test, Shoulder Flexibility, Goniometry	2	2
	4.3	Dynamic Mobility Testing (Functional Reach, Y-Balance)	2	2
	4.4	Posture and Alignment Screening (Posture Grid, Plumb Line)	3	3
	4.5	Functional Movement Screening (FMS) Basics	3	4
5		**TEACHER SPECIFIC CONTENT**		

<table>
<tr><td rowspan="3">Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)</td></tr>
<tr><td>Mode of Assessment</td></tr>
<tr><td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
● Lesson plans
● Group Activities</td></tr>
<tr><td rowspan="2">Mode of Assessment</td><td>Continues Comprehensive Assessment (CCA)
Total Mark - 30
<table><tr><th>Assessment Method (CCA)</th><th>Marks</th></tr><tr><td>Assignment</td><td>15</td></tr><tr><td>Individual Involvement & Viva</td><td>15</td></tr><tr><td>Total</td><td>30</td></tr></table></td></tr>
<tr><td>End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 70
Duration – 3 hrs
<table><tr><th>Assessment Method (ESE)</th><th>Marks</th></tr><tr><td>Lesson Plan</td><td>30</td></tr><tr><td>Demonstration & Presentation</td><td>30</td></tr><tr><td>Viva</td><td>10</td></tr><tr><td>Total</td><td>70</td></tr></table></td></tr>
</table>

References

1. American College of Sports Medicine. (2022). ACSM's health-related physical fitness assessment manual (6th ed., pp. 120–140). Wolters Kluwer.
2. Baechle, T. R., & Earle, R. W. (2022). NSCA’s essentials of personal training (3rd ed., pp. 285–305). Human Kinetics.
3. Beekley, M. D. (2022). Principles of exercise testing and interpretation (6th ed., pp. 210–230). Springer.
4. Coburn, J. W., & Malek, M. H. (2020). NSCA’s essentials of sport science (pp. 200–220). Human Kinetics.

Suggested Readings

1. Heyward, V. H., & Gibson, A. (2021). Advanced fitness assessment and exercise prescription (8th ed., pp. 150–170). Human Kinetics.
2. Howley, E. T., & Franks, B. D. (2021). Health fitness instructor's handbook (6th ed., pp. 310–330). Human Kinetics.
3. Pescatello, L. S. (2021). ACSM's guidelines for exercise testing and prescription (11th ed., pp. 195–215). Wolters Kluwer.
4. Thompson, W. R. (2021). Fitness professional's handbook (8th ed., pp. 235–255). Human Kinetics.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training
Programme	B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training
Course Name	Self Defense Training and Fitness
Type of Course	MPC
Course Code	MG3MPCMFT200
Course Level	200
Course Summary	This course introduces students to the fundamentals of martial arts with a focus on Taekwondo, Karate, and Krav Maga, integrating both traditional skills and modern self-defense applications. Students will develop physical fitness, mental discipline, and situation-based self-defense techniques through theoretical and practical components. Emphasis will be placed on improving reflexes, flexibility, situational awareness, and confidence. The course bridges martial arts with fitness goals and equips students with tools for real-life personal safety and self-control.

<table>
<tr><td>Semester</td><td>3</td><td colspan="2">Credits</td><td>4</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>75</td></tr>
<tr><td>Pre-requisites, if any</td><td colspan="5">Students must have basic physical fitness and body coordination awareness. Basic knowledge of movement mechanics is desirable.</td></tr>
</table>

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	**Demonstrate fundamental techniques of Taekwondo, Karate, and Krav Maga.**	**K**	1,2,3
CO2	Apply basic self-defense strategies in various real-life threat scenarios.	**A**	2
CO3	Evaluate and improve personal fitness through martial arts training.	**C**	2,3
CO4	Analyze the mental and psychological aspects of self-defense.	**AN**	2,3,10
CO5	Design basic martial arts-based fitness routines for self and others.	**U**	2,3,4,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	1	1	2	0	0	0	0	0	0	0
CO2	**0**	2	0	0	0	0	0	0	0	0
CO3	0	**1**	3	0	0	0	0	0	0	0
CO4	0	2	**2**	0	0	0	0	0	0	1
CO5	0	1	1	3	0	0	0	0	0	3
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Introduction to Martial Arts & Self-Defense**		
	1.1	History and evolution of martial arts	4	1
	1.2	Overview of Taekwondo, Karate, and Krav Maga	4	2
	1.3	Principles of self-defense and ethics in martial arts	3	1
	1.4	Martial arts as a fitness regime	4	2
	1.5	Legal implications and use-of-force awareness	4	2
2	2	**Taekwondo – Foundation and Applications**		
	2.1	Basic stances, blocks, punches, and kicks		1
	2.2	Forms (Poomsae) and combinations	4	2
	2.3	Sparring techniques and pad drills	4	2
	2.4	Fitness drills using Taekwondo movements	5	3
	2.5	Self-defense using Taekwondo principles	5	1
3	3	**Karate – Techniques and Conditioning**		
	3.1	Historical Background of Karate	4	3
	3.2	Basic Movements, Punches (Tsuki), Blocks (Uke), Kicks (Geri)	4	3
	3.3	Kihons – (Fundamental Techniques) Kata (Forms) and Kumite (Sparring)	4	2
	3.4	Karate for Cardiovascular Endurance and Reflex Development	5	3
	3.5	Meditation, Breathing Techniques, and Mental fitness	5	3
4	4	**Krav Maga – Tactical Self Defense**		
	4.1	Krav Maga Stance, Rolling techniques	3	2
	4.2	Defense against grabs, chokes, and punches	3	2
	4.3	Situational training: attacks from front, back, or while on ground	3	4
	4.4	Defense against weapons (knife/stick/basic simulation)	3	4
	4.5	Stress training and verbal de-escalation	4	2
5	5	**TEACHER SPECIFIC CONTENT**		

<table>
<tr>
<td rowspan="3">Teaching and Learning Approach</td>
<td>Classroom Procedure (Mode of transaction)</td>
</tr>
<tr>
<td>Mode of Assessment</td>
</tr>
<tr>
<td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
● Lesson plans
● Hands on training</td>
</tr>
<tr>
<td rowspan="2">Mode of Assessment</td>
<td>Continues Comprehensive Assessment (CCA)
Total Mark - 30
<table>
<tr><th>Assessment Method (CCA)</th><th>Marks</th></tr>
<tr><td>Assignment</td><td>15</td></tr>
<tr><td>Individual Involvement & Viva</td><td>15</td></tr>
<tr><td>Total</td><td>30</td></tr>
</table>
</td>
</tr>
<tr>
<td>End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 70
Duration – 3 hrs
<table>
<tr><th>Assessment Method (ESE)</th><th>Marks</th></tr>
<tr><td>Lesson Plan</td><td>30</td></tr>
<tr><td>Demonstration & Presentation</td><td>30</td></tr>
<tr><td>Viva</td><td>10</td></tr>
<tr><td>Total</td><td>70</td></tr>
</table>
</td>
</tr>
</table>

References

1. Kim, S. (2023). Taekwondo techniques & training (pp. 125–145). Human Kinetics.
2. Lee, B. (2022). The Tao of martial arts (pp. 75–90). Dragon Door Publications.
3. Levine, D., & Yodice, R. (2021). Krav Maga for beginners: A step-by-step guide (pp. 95–110). Ulysses Press.
4. Lichtenfeld, I., & Yanilov, E. (2020). Krav Maga: How to defend yourself against armed assault (pp. 140–160). Black Belt Books.

Suggested Readings

1. Moyers, S. (2023). Mental toughness in martial arts training (pp. 60–80). Fitness Fighters Press.
2. Nakamura, H. (2021). Modern Karate-Do (pp. 100–120). Kodansha International.
3. Smith, R. (2022). Martial arts for mind and body: Self-defense and spiritual fitness (pp. 130–150). Routledge.
4. Yamamoto, K. (2021). Applied karate and self-defense (pp. 110–125). Martial Arts Press

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Traditional Sports in Kerala**
Type of Course	**MDC**
Course Code	**MG3MDCMFT200**
Course Level	200
Course Summary	This course introduces students to the rich heritage of traditional sports and indigenous games of Kerala. It explores their cultural roots, physical benefits, historical evolution, and relevance in today's fitness and recreation programs. The course aims to provide students with conceptual clarity and appreciation of these sports, as well as how they contribute to physical health, teamwork, and social bonding. The content is designed to be simple, descriptive, and student-friendly to support exam preparation and broader understanding.

<table>
<tr><td>Semester</td><td>3</td><td colspan="2">Credits</td><td>3</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>45</td></tr>
<tr><td>Pre-requisites, if any</td><td colspan="5">Basic interest in Kerala culture and physical activities (no prior knowledge required), Students from all disciplines (sports, fitness, social sciences, tourism, education, etc.) can join, Ability to read and understand simple Malayalam terms may be helpful (not mandatory)</td></tr>
</table>

COURSE OUTCOMES (CO)

CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	**Identify and describe various traditional sports and games of Kerala.**	K	1,2,10
CO2	Understand the historical, social, and cultural background of Kerala's indigenous sports.	K	2,4,5
CO3	Explain the rules, equipment, and format of selected traditional games.	K	2,4
CO4	Analyze the fitness and physical benefits of traditional sports activities.	U	2,4,5
CO5	Promote awareness and participation of traditional sports in community and educational settings.	U	2,4,10

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	1	1	0	0	0	0	0	0	0	3
CO2	0	1	0	1	2	0	0	0	0	0
CO3	0	2	0	2	0	0	0	0	0	0
CO4	0	2	0	3	1	0	0	0	0	0
CO5	0	2	0	1	0	0	0	0	0	2
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Introduction to Traditional Sports and Kerala's Heritage**		
	1.1	Definition and Importance of Traditional Sports	5	1
	1.2	Historical Overview of Sports in Kerala	4	2
	1.3	Festivals and Traditional Sports (Onam, Pooram, etc.)	4	2
	1.4	Role of Traditional Sports in Community and Culture	4	3
2	2	**Indigenous Games of Kerala**		
	2.1	Kilithattu Kali, Goli Kali, Kuttiyum Kolum	4	1
	2.2	Vadamvali (Tug of War), Uriyadi, Chadangu Kali	4	1
	2.3	Indoor Traditional Games – Pallankuzhi, Kacha Kali, etc.	4	2
	2.4	Local Variants – Tribal and Regional Games of Kerala	5	2
3	3	**Martial Art-Based Traditional Sports**		
	3.1	Kalaripayattu – Origin, Training, and Competitions	4	2
	3.2	Velakali, Paricha Kali – Ritual & Symbolic Combat	3	1
	3.3	Use of Traditional Weapons (stick fighting, sword, etc.)	2	2
	3.4	Physical Fitness and Discipline in Martial Art Forms	2	3
4		**TEACHER SPECIFIC CONTENT**		

<table>
<tr><td rowspan="3">Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
● Lesson plans
● Hands on training</td></tr>
<tr><td>Mode of Assessment</td></tr>
<tr><td>MODE OF ASSESSMENT Continuous Comprehensive Assessment (CCA)
CCA for Theory :25 Marks
CCA -10 marks (Assignment)
CCA -15 mark, (Viva, Presentation, individual involvement)</td></tr>
<tr><td>Assessment Type</td><td>End Semester Examination (ESE)

ESE Theory –50 marks (Written examination theory)
Duration of Examination 1.5 hrs

Different part of examinations –

Part 1, MCQ 10x1=10 (number of questions 12-number of questions to be answered 10)
Part 2 - Short Answer – marks - 5x2=10 (number of questions 7 -number of questions to be answered 5)
Part 3 - Short Essay - 6x5=30 (number of questions 8-number of questions to be answered 6)</td></tr>
</table>

References

1. Indian Ministry of Youth Affairs & Sports. (2021). Revival of indigenous sports in India: A policy review (pp. 95–115). Government Publication.
2. Joseph, L. (2023). Kalaripayattu and indigenous training systems in Kerala (pp. 130–150). Orient BlackSwan.
3. Kumar, P. S. (2020). Heritage sports and physical culture of South India (pp. 180–200). Notion Press.
4. Menon, A. (2021). Folk sports of Kerala: A cultural study (pp. 145–165). DC Books.
5. Nair, R. G. (2022). Traditional games and martial arts of Kerala (pp. 100–120). Calicut University Press.

Suggested Readings

1. Singh, R., & Devi, M. (2023). Traditional Indian physical activities: A regional perspective (pp. 210–230). Atlantic Publishers.
2. Thomas, R. (2021). Cultural traditions of Kerala through games and rituals (pp. 160–180). Sage Publications.
3. Varghese, M. (2022). Reclaiming traditional sports for modern fitness (pp. 85–105). Zorba Books.

SEMESTER IV

Semester 4

Course Code	Title of the Course	Type of the Course	Credit	Hours /week	Hour Distribution /week		
					L	P	O
MG4SDCMFT200	Diet and Nutrition in Fitness Programming	SDC	4	4	4	0	0
MG4SDCMFT201	Martial Arts for Fitness & Self Defense - II - Judo & Kurash, Karate	SDC	4	5	3	2	0
MG4SDCMFT202	Principles of injury assessment and Safety planning	SDC	4	5	3	2	0
MG4MPCMFT200	Yoga and Life Style Management	MPC	4	5	3	2	0
MG4SECMFT200	Principles of Fitness Management	SEC	3	3	3	0	0
MG4VACMFT200	Soft Skills for Fitness Professionals	VAC	3	3	3	0	0
MG4INTMFT200	Internship	INT	2	60 HOURS			

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Mahatma Gandhi University Kottayam

Faculty/ Discipline	Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training
Programme	B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training
Course Name	Diet and Nutrition in Fitness Programming
Type of Course	SDC
Course Code	MG4SDCMFT200
Course Level	200
Course Summary	This course introduces students to the core principles of diet and nutrition with a focus on fitness and performance. It explores the role of macronutrients, micronutrients, hydration, supplementation, and meal timing in relation to exercise programming. Students will learn to assess dietary needs and design customized diet plans that align with individual fitness goals. By the end of the course, learners will be equipped with knowledge to support physical training and enhance performance through optimal nutrition strategies.

Semester	4	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		4	0	0	60
Pre-requisites, if any	Basic understanding of human physiology, physical fitness, and energy metabolism (from prior courses such as Principles of Physical Fitness & Wellbeing and Human Body: Structure to Functioning & Energy Metabolism).				

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	**Understand the role and importance of nutrition in physical fitness and performance.**	**U**	1,2,3
CO2	Identify the nutritional requirements of individuals based on activity levels and fitness goals.	K	**1,2,4,5**
CO3	Design balanced diet plans and evaluate nutritional quality.	A	**2,4,6**
CO4	Analyze the role of supplementation and hydration in athletic training.	An	**3,4,6**
CO5	Apply the principles of sports nutrition to support endurance, strength, and recovery.	A	**2,4,5**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO AR ICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	1	2	1	0	0	0	0	0	0	0
CO2	**2**	2	0	1	1	0	0	0	0	0
CO3	0	1	0	2	0	3	0	0	0	0
CO4	0	0	2	2	0	2	0	0	0	0
CO5	0	2	0	1	3	0	0	0	0	0

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Module	Units	Course description	Hrs	CO No.
1	1	**Fundamentals of Nutrition**		
	1.1	Introduction to Nutrition and Health	3	2
	1.2	Classification and Functions of Nutrients	3	2
	1.3	Role of Carbohydrates, Proteins, and Fats	3	3
	1.4	Importance of Vitamins and Minerals	4	2
	1.5	Energy Balance and Caloric Needs	4	4
2	2	**Nutrition for Fitness and Training**		
	2.1	Nutritional Demands for Different Exercise Types (Endurance vs. Strength)	3	2
	2.2	Pre-Workout and Post-Workout Nutrition	3	4
	2.3	Meal Timing and Frequency	4	4
	2.4	Glycogen Storage and Recovery	4	3
	2.5	Case Study: Diet Plans for Runners, Lifters, and Group Trainers	3	2
3	3	**Designing a Balanced Diet Plan**		
	3.1	Understanding RDA (Recommended Dietary Allowance)	3	2

	3.2	Principles of Meal Planning and Portion Control	3	2
	3.3	Reading Food Labels and Calculating Calories	2	1
	3.4	Macronutrient Ratios for Different Fitness Goals	3	3
	3.5	Creating Weekly Diet Charts for General and Athletic Population	3	3
4	4	**Special Considerations in Nutrition**		
	4.1	Hydration and Electrolyte Balance	2	3
	4.2	Vegetarian, Vegan, and Cultural Diet Adaptations	2	3
	4.3	Managing Nutrition for Weight Loss and Muscle Gain	2	4
	4.4	Food Intolerances and Allergies	3	4
	4.5	Nutrition for Special Populations: Children, Elderly, Pregnant Athletes	3	1
5		**TEACHER SPECIFIC CONTENT**		

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Lecture (Power Point presentation) ● Demonstration ● Peer teaching ● Lesson plans ● Hands on training	
	Mode of Assessment	
	MODE OF ASSESSMENT Continuous Comprehensive Assessment (CCA) CCA for Theory:30 Marks CCA -15 marks (Assignment, viva) CCA -15 mark, (Presentation, individual involvement)	
Assessment Type	**End Semester Examination (ESE)** **ESE Theory –70 marks (Written examination theory)** **Duration of Examination 2 hrs** **Different part of examinations** – **Part 1**, MCQ 10x1=10 (number of questions 12-number of questions to be answered 10) **Part 2** - Short Answer – marks - 5x2=10 (number of questions 7 -number of questions to be answered 5) **Part 3 -** Short Essay - 4x5=20 (number of questions 6-number of questions to be answered 4) **Part 4** - Essay-2 x 15=30 - (number of questions 3-number of questions to be answered 2).	

References

1. Antonio, J., & Kalman, D. (2021). Essentials of sports nutrition and supplements (2nd ed., pp. 305–325). Springer.
2. Benardot, D. (2020). Advanced sports nutrition (3rd ed., pp. 185–205). Human Kinetics.
3. Dunford, M., & Doyle, J. A. (2023). Nutrition for sport and exercise (4th ed., pp. 220–240). Cengage Learning.
4. Jeukendrup, A., & Gleeson, M. (2018). Sport nutrition: An introduction to energy production and performance (3rd ed., pp. 145–165). Human Kinetics.
5. Kerksick, C. M., & Fox, C. (2021). NSCA's guide to sport and exercise nutrition (2nd ed., pp. 110–130). Human Kinetics.

Suggested Readings

1. Lunn, W. R., & Borsheim, E. (2022). Nutrition for sport, exercise, and health (2nd ed., pp. 310–330). Wolters Kluwer.
2. Manore, M. M., Thompson, J., & Vaughan, L. A. (2022). Sport nutrition for health and performance (4th ed., pp. 275–295). Human Kinetics.
3. Maughan, R. J. (Ed.). (2022). The encyclopaedia of sports medicine: Sports nutrition (Vol. XIX, pp. 200–220). Wiley-Blackwell.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training				
Programme	B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training				
Course Name	Martial Arts for Fitness & Self Defense – II: Judo, Kurash & Karate				
Type of Course	SDC				
Course Code	MG4SDCMFT201				
Course Level	200				
Course Summary	This course builds upon foundational martial arts by introducing students to Judo, Kurash, and Karate—three dynamic combat arts that contribute significantly to physical and mental conditioning. Through theoretical instruction and hands-on practice, learners will understand the role of these martial arts in fitness development, injury prevention, self-defense, and overall wellness. Emphasis is placed on discipline, movement patterns, balance, coordination, and cultural context. The course provides structured progression in each style with real-world application for health and fitness programming.				
Semester	4	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	75
Pre-requisites, if any	Basic understanding of martial arts, fitness, and self-defense principles.				

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	**Demonstrate fundamental techniques of Judo, Kurash, and Karate.**	K	1,2,5
CO2	Understand the physical and psychological benefits of each martial art in fitness training.	A	2,5
CO3	Apply martial arts drills to improve strength, flexibility, endurance, and coordination.	U	2,4,5
CO4	Integrate martial arts concepts for personal safety and self-defense training.	U	4,5,6
CO5	Develop martial-arts-based fitness modules suitable for group and individual programs.	A	2,4,5

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**
CO 1	2	2	0	0	2	0	0	0	0	0
CO2	0	3	**0**	**0**	3	0	0	0	0	0
CO3	0	2	0	2	**1**	0	0	0	0	0
CO4	0	0	0	2	2	1	0	0	0	0
CO5	0	3	0	1	3	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Introduction to Martial Arts in Fitness Training**		
	1.1	Evolution and Philosophy of Martial Arts	4	1
	1.2	Role of Martial Arts in Modern Fitness Programs	4	1
	1.3	Psychological Benefits: Focus, Discipline, Confidence	4	2
	1.4	Principles of Balance, Coordination, Agility, and Core Strength	5	2
	1.5	Safety Measures and Injury Prevention in Martial Arts Practice	5	1
2	2	**Basics of Judo**		
	2.1	History and Origin of Judo	4	1
	2.2	Dojo Etiquette and Terminology	4	1
	2.3	Fundamental Techniques: Kuzushi (Balance Breaking), Ukemi (Falls)	5	3
	2.4	Basic Throws: Osoto Gari, O Goshi, Ippon Seoi Nage	5	2
	2.5	Judo for Fitness: Strength, Flexibility, and Reaction Training	4	2
3	3	**Basics of Kurash**		

	3.1	History and Origin of Kurash (Uzbek Traditional Sport)	4	2
	3.2	Kurash Rules and Techniques vs. Judo Comparison	4	3
	3.3	Common Throws and Grips	3	2
	3.4	Conditioning Drills Using Kurash Movement Patterns	4	2
	3.5	Fitness Elements in Kurash: Agility, Power, and Mobility	4	3
4	4	**Basics of Karate**		
	4.1	Historical Background of Karate	2	4
	4.2	Basic Movements, Punches (Tsuki), Blocks (Uke), Kicks (Geri)	2	1
	4.3	Kihons – (Fundamental Techniques) Kata (Forms) and Kumite (Sparring)	2	3
	4.4	Karate for Cardiovascular Endurance and Reflex Development	3	2
	4.5	Meditation, Breathing Techniques, and Mental fitness	3	2
5		**TEACHER SPECIFIC CONTENT**		

<table>
<tr>
<td rowspan="3">Teaching and Learning Approach</td>
<td>Classroom Procedure (Mode of transaction)</td>
</tr>
<tr>
<td>Mode of Assessment</td>
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<td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
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● Group Activities</td>
</tr>
<tr>
<td rowspan="2">Mode of Assessment</td>
<td>Continues Comprehensive Assessment (CCA)
Total Mark - 30
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<tr><th>Assessment Method (CCA)</th><th>Marks</th></tr>
<tr><td>Assignment</td><td>15</td></tr>
<tr><td>Individual Involvement & Viva</td><td>15</td></tr>
<tr><td>Total</td><td>30</td></tr>
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</tr>
<tr>
<td>End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 70
Duration – 3 hrs
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<tr><th>Assessment Method (ESE)</th><th>Marks</th></tr>
<tr><td>Lesson Plan</td><td>30</td></tr>
<tr><td>Demonstration & Presentation</td><td>30</td></tr>
<tr><td>Viva</td><td>10</td></tr>
<tr><td>Total</td><td>70</td></tr>
</table>
</td>
</tr>
</table>

References

1. Gentry, C. (2021). The complete martial arts training manual: An integrated approach (pp. 110–130). Tuttle Publishing.
2. Gifford, C. (2020). Martial arts: The ultimate guide to styles, stances, and moves (pp. 85–105). Arcturus Publishing.
3. Hamidov, R. (2023). Kurash: Traditional wrestling and modern fitness (pp. 140–160). Tashkent Sport Press.
4. Kano, J., & Watson, B. (2019). Kodokan Judo: The essential guide to Judo by its founder (pp. 95–115). Kodansha International.

Suggested Readings

1. Nakayama, M. (2021). Dynamic Karate: Instruction by the master (pp. 150–170). Kodansha.
2. Nishiyama, H., & Brown, R. C. (2023). Karate: The art of "empty hand" fighting (pp. 200–220). Tuttle Publishing.
3. Rebek, K., & Green, T. A. (2022). Martial arts of the world: An encyclopedia of history and innovation (Vol. 1, pp. 250–270). ABC-CLIO.
4. Wilk, J. E., & McNally, L. D. (2020). Combat conditioning: Martial arts training for strength and performance (pp. 180–200). Human Kinetics.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Principles of Injury Assessment and Safety Planning**
Type of Course	**SDC**
Course Code	**MG4SDCMFT202**
Course Level	200
Course Summary	This course introduces students to the fundamental principles and practices of injury assessment, prevention, and management within the context of sports and fitness. It covers the types and causes of common fitness-related injuries, principles of first aid, emergency response protocols, and safety planning strategies. The practical component reinforces hands-on skills in injury evaluation, basic taping, CPR, and safe exercise planning. The course aims to build competence in responding to injuries confidently and safely, preparing students for careers in personal training, coaching, or fitness instruction.

Semester	4	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	Students should have completed foundational courses in anatomy, physiology, and fitness training concepts.				

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	**Identify and classify common fitness-related injuries and understand their mechanisms.**	**K**	1
CO2	Perform basic injury assessments and implement initial management strategies.	**A**	**2,4,5**

CO3	Apply first aid principles and demonstrate emergency response techniques.	**C**	**2,3,5**
CO4	Develop safety plans and risk management strategies for fitness environments.	**S**	**3,5,6**
CO5	Understand preventive techniques and ergonomic adjustments to reduce injury risk.	**U**	**2,4,5**
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**
CO 1	**1**	0	0	0	0	0	0	0	0	0
CO2	0	3	0	1	3	0	0	0	0	0
CO3	0	2	2	0	**1**	0	0	0	0	0
CO4	0	0	2	0	2	**1**	0	0	0	0
CO5	0	3	0	1	3	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Introduction to Injury and Safety Concepts**		
	1.1	Understanding Sports and Fitness-Related Injuries	4	3
	1.2	Classification of Injuries (Acute vs. Chronic)	4	1
	1.3	Mechanism of Injury: Intrinsic and Extrinsic Factors	4	4
	1.4	Overview of Injury Prevention Strategies	5	2
	1.5	Safety Protocols in Fitness Training Environments	5	2
2	2	**Basic Injury Assessment**		
	2.1	Principles of Injury Evaluation (SOAP Approach)	4	1
	2.2	Understanding Signs and Symptoms	4	1
	2.3	On-field and Off-field Assessment Techniques	5	1

	2.4	Range of Motion and Strength Testing	5	2
	2.5	When to Refer: Recognizing Red Flags	4	2
3	3	**Common Injuries and Their Management**		
	3.1	Musculoskeletal Injuries: Sprains, Strains, Dislocations	4	2
	3.2	Overuse Injuries: Tendinitis, Stress Fractures	4	2
	3.3	Head, Neck, and Spinal Injuries – Precautionary Handling	3	3
	3.4	Cardiovascular Emergencies in Fitness (e.g., Hypertensive Episodes)	4	
	3.5	Immediate Care and Rehabilitation Protocols	4	2
4	4	**Principles of First Aid and Emergency Planning**		
	4.1	First Aid: Definition, Objectives, and Scope	2	1
	4.2	The "DRABC" of Emergency Response	2	5
	4.3	CPR and AED Application (Basic Certification Practice)	2	2
	4.4	Bandaging, Taping, and Splinting Techniques	3	3
	4.5	Emergency Evacuation Procedures and Communication Plans	3	3
5		**TEACHER SPECIFIC CONTENT**		

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Lecture (Power Point presentation) ● Demonstration ● Peer teaching ● Lesson plans ● Group Activities
Mode of Assessment	**Continues Comprehensive Assessment (CCA)** Total Mark - 30 Assessment Method (CCA) – Marks Assignment – 15 Individual Involvement & Viva – 15 Total – 30
	End Semester Examination (ESE) Assessment Method – Practical Examination Total Marks – 70 Duration – 3 hrs Assessment Method (ESE) – Marks Lesson Plan – 30 Demonstration & Presentation – 30 Viva – 10 Total – 70

Assessment Method (CCA)	Marks
Assignment	15
Individual Involvement & Viva	15
Total	30

Assessment Method (ESE)	Marks
Lesson Plan	30
Demonstration & Presentation	30
Viva	10
Total	70

References

1. American College of Sports Medicine. (2021). ACSM's guidelines for exercise testing and prescription (11th ed., pp. 210–230). Wolters Kluwer.
2. Arnheim, D. D., & Prentice, W. E. (2021). Principles of athletic training: A competency-based approach (17th ed., pp. 365–385). McGraw-Hill.
3. Baechle, T., & Earle, R. W. (2022). NSCA's essentials of strength training and conditioning (5th ed., pp. 150–170). Human Kinetics.
4. Maud, P. J., & Foster, C. (2020). Physiological assessment of human fitness (2nd ed., pp. 110–130). Human Kinetics.

Suggested Readings

1. McGill, S. (2020). Back mechanic: The step-by-step guide to fix your back pain (3rd ed., pp. 160–180). Backfitpro Inc.
2. Pesterfield, S. (2023). Sports first aid and injury prevention (2nd ed., pp. 95–115). Human Kinetics.
3. Prentice, W. E. (2023). Essentials of athletic injury management (12th ed., pp. 240–260). McGraw-Hill Education.
4. Starkey, C. (2022). Examination of orthopedic and athletic injuries (5th ed., pp. 310–330). F.A. Davis Company.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Yoga and Life Style Management**
Type of Course	**MPC**
Course Code	**MG4MPCMFT200**
Course Level	200
Course Summary	This course introduces students to the philosophy, practice, and modern applications of Yoga for healthy living. It helps students understand the importance of yoga in daily life, how it affects physical health, mental peace, emotional stability, and lifestyle habits. Through regular theory and practical sessions, students learn to create a balanced, stress-free, and mindful lifestyle, empowering them to manage academic and personal life more efficiently.

Semester	4	**Credits**		4	**Total Hours**
Course details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	**75**
Pre-requisites, if any	Students must have a basic awareness of physical activity and a willingness to explore body-mind practices. No prior experience in yoga is required.				

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	**Understand the fundamental principles and philosophy of yoga.**	**K**	1
CO2	Demonstrate basic yoga postures, breathing techniques, and relaxation methods.	**U**	2
CO3	Apply yogic practices for stress management and lifestyle correction.	**U**	2
CO4	Evaluate their current lifestyle and implement yoga-based improvements.	**E**	2
CO5	Create simple daily yoga routines for holistic wellness.	**C**	2

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	0	0	0	0	0	0	0	0	0
CO2	0	2	0	0	0	0	0	0	0	0
CO3	0	1	0	0	0	0	0	0	0	0
CO4	0	3	0	0	0	0	0	0	0	0
CO5	0	2	0	0	0	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Foundations of Yoga and Lifestyle**		
	1.1	Definition and objectives of yoga	2	1
	1.2	History and schools of yoga	2	2
	1.3	Concept of lifestyle management	2	2
	1.4	Modern lifestyle disorders and their causes	3	1
	1.5	Role of yoga in holistic health	3	4
2	2	**Asanas for Daily Wellness**		
	2.1	Standing, sitting, prone and supine postures	4	1
	2.2	Joint mobility and warm-up practices	4	1
	2.3	Postures for back pain, digestion, and flexibility	5	2
	2.4	Benefits and precautions of different asanas	5	4
	2.5	Designing a beginner-level asana sequence	4	2
3	3	**Pranayama and Breath Management**		
	3.1	Basic pranayama techniques (Anulom-Vilom, Bhramari, Sheetali)	4	2
	3.2	Importance of breath in health and stress	4	2
	3.3	Energy channels (Nadis) and their purification	3	3
	3.4	Breathing for relaxation and sleep improvement	4	3
	3.5	Contraindications and precautions	4	3
4	4	**Meditation, Relaxation, and Stress Management**		
	4.1	Mindfulness and concentration practices	4	2
	4.2	Yogic meditation techniques (Om chanting, Trataka)	4	1
	4.3	Deep relaxation techniques (Yoga Nidra)	4	2
	4.4	Stress, anxiety, and emotional balance through yoga	5	2
	4.5	Integrating meditation into daily life	5	5
5		**TEACHER SPECIFIC CONTENT**		

<table>
<tr><td rowspan="2">Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
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● Hands on training</td></tr>
<tr><td>Mode of Assessment

Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
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Total Mark - 30

<table><tr><th>Assessment Method (CCA)</th><th>Marks</th></tr><tr><td>Assignment</td><td>15</td></tr><tr><td>Individual Involvement & Viva</td><td>15</td></tr><tr><td>Total</td><td>30</td></tr></table></td></tr>
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Assessment Method – Practical Examination
Total Marks – 70
Duration -3 hrs

<table><tr><th>Assessment Method (ESE)</th><th>Marks</th></tr><tr><td>Lesson Plan</td><td>30</td></tr><tr><td>Demonstration & Presentation</td><td>30</td></tr><tr><td>Viva</td><td>10</td></tr><tr><td>Total</td><td>70</td></tr></table></td></tr>
</table>

References

1. Bhavanani, A. B. (2022). Understanding yoga for modern life (pp. 90–105). Dhivyananda Publications.
2. Chopra, D. (2023). Total meditation: Practices in living the awakened life (pp. 120–135). Harmony.
3. Desikachar, T. K. V. (2020). The heart of yoga: Developing a personal practice (pp. 145–160). Inner Traditions.
4. Iyengar, B. K. S. (2021). Light on life: Yoga journey to wholeness (pp. 180–200). Rodale Books.
5. Nambiar, S. (2022). Yoga for stress-free life (pp. 75–90). Penguin India.

Suggested Readings

1. Saraswati, S. (2022). Asana pranayama mudra bandha (4th ed., pp. 210–230). Yoga Publications Trust.
2. Satyananda, S. (2023). Yoga nidra for lifestyle management (pp. 95–115). Yoga Publications Trust.
3. Tiwari, M. (2021). Science of yoga and lifestyle management (pp. 130–150). Lotus Press.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc .(Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Principles of Fitness Management**
Type of Course	**SEC**
Course Code	**MG4SECMFT200**
Course Level	200
Course Summary	This course introduces students to the fundamental principles of managing fitness facilities and services. It focuses on the organizational, operational, and strategic aspects of running a health club or sports fitness center. Topics include understanding fitness industry structure, human resource management, facility operations, marketing, customer service, ethics, and sustainability. Designed for vocational students in multi-sports and fitness training, the course emphasizes practical applications, critical thinking, and ethical practices for successful career growth in the fitness industry.

Semester	4	**Credits**		3	**Total Hours**
Course details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	0	0	**45**
Pre-requisites, if any	Basic interest in Kerala culture and physical activities (no prior knowledge required), Students from all disciplines (sports, fitness, social sciences, tourism, education, etc.) can join, Ability to read and understand simple Malayalam terms may be helpful (not mandatory)				

COURSE OUTCOMES (CO)

CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	Describe the core principles and functions of fitness management.	R	1,2
CO2	Apply basic management techniques to health clubs and fitness operations.	A	3,4
CO3	Analyze challenges in fitness facility administration and suggest solutions.	U	1
CO4	Communicate effectively with clients and staff in fitness business contexts.	A	3,5

CO5	Demonstrate awareness of ethics, sustainability, and inclusiveness in fitness management.	U	5
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2	0	0	0	0	0	0	0	0
CO 2	0	0	2	3	0	0	0	0	0	0
CO 3	1	0	0	0	0	0	0	0	0	0
CO4	0	0	2	0	1	0	0	0	0	0
CO5	0	0	0	0	2	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	Introduction to Fitness Management		
	1.1	Meaning and Scope of Fitness Management Definition, objectives, and importance, Fitness industry in India and global trends	5	1
	1.2	Principles of Management in Fitness Settings, Planning, organizing, leading, controlling, Role of a fitness manager	5	2
	1.3	Types of Fitness Facilities and Services, Health clubs, gyms, boutique studios, sports centers, Public vs. private sector fitness services	5	1
2	2	Operations and Human Resource Management		
	2.1	Facility Operations and Safety, Equipment management, maintenance schedules, Safety protocols and legal considerations	5	2
	2.2	Human Resource Management in Fitness, Recruitment, training, and evaluation of staff, Motivation and teamwork	5	2
	2.3	Customer Service in Fitness Facilities, Client onboarding, complaint handling, Enhancing client experience	5	1
3	3	Marketing, Ethics, and Sustainability		
	3.1	Basics of Marketing for Fitness Businesses, Branding, advertising, and promotions, Social media marketing for gyms and trainers	5	2
	3.2	Ethics and Professional Conduct in Fitness, Code of conduct, privacy, and professional boundaries, Inclusiveness and equity in fitness services	5	2
	3.3	Sustainability in Fitness Management, Eco-friendly operations, Community engagement and social responsibility	5	2
4		**TEACHER SPECIFIC CONTENT**		

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** **• Class Room Lectures** **• Case studies and Discussions** **• Presentation Practices** **Mode of Assessment** **MODE OF ASSESSMENT** **Mode of Assessment** **Interactive Lectures** **Group discussions**
Assessment Type	**Assessment Methods** **A. Continuous Comprehensive Assessment (CCA)** **Theory - 25** **Assessment Methods** **Class test/unit test** **Assignment/case study** **End Semester Evaluation (ESE)** **Theory-50 Marks** **Duration of Examination – 1.5 hrs.** **Pattern of examination for Theory - Non-MCQ Different parts of written examination** **Question Type Number of questions to be answered Answer word/page limit Marks** **Part A- MCQ 10 out of 12 Choose the correct answer 10*1=10** **Part B 10 out of 12 Very Short Answers 10*2=20** **Part C 4 out of 6 Short Answers 4*5=20**

References

1. Bedi, M. (2019). *Management of sports and physical education*. Friends Publications India.
2. Dhall, M. (2018). *Fitness training and management*. Sports Publication.
3. Singh, A., & Rathee, N. (2015). *Management of health and fitness*. Khel Sahitya Kendra.
4. Kumar, S. (2017). *Sports management and administration*. Sports Publication.
5. Sharma, V. M. (2016). *Principles of management in physical education and sports*. Khel Sahitya Kendra.

Suggested Readings

1. Kaur, R. (2020). *Sports facility planning and management*. Friends Publications India.
2. Sharma, A., & Bansal, R. (2019). *Fitness and wellness management*. Khel Sahitya Kendra.
3. Bhardwaj, A. (2017). *Organization and administration in physical education*. Friends Publications India.
4. Jain, A. (2021). *Fundamentals of sports marketing*. Sports Publication.
5. Choudhary, R. (2018). *Professional ethics in sports and fitness*. Friends Publications India.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc .(Honours) Advanced Course in Multi Sport and Fitness Training**
Course Name	**Soft Skills for Fitness Professionals**
Type of Course	**VAC**
Course Code	**MG4VACMFT200**
Course Level	200
Course Summary	This course aims to develop the communication abilities, interpersonal skills, and professional behavior needed to succeed in the fitness industry. Students will learn effective ways to interact with clients, build confidence, handle conflicts, and maintain ethical conduct in health clubs and personal training environments. The course also covers teamwork, leadership, and cultural sensitivity to help students work effectively with diverse clients and colleagues.

Semester	4	**Credits**		3	**Total Hours**
Course details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	0	0	**45**
Pre-requisites, if any	Basic communication skills, willingness for self-reflection, familiarity with teamwork, and basic computer literacy.				

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	Explain the importance of soft skills in the fitness industry.	U	1,10
CO2	Demonstrate effective verbal and non-verbal communication in professional fitness settings.	A	2,7
CO3	Apply interpersonal skills for client relationship management.	A	3,4
CO4	Exhibit leadership and teamwork abilities in group fitness and club operations.	An	1
CO5	Practice ethical behavior, inclusiveness, and cultural sensitivity in client dealings.	A	1

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	0	0	0	0	0	0	0	0	3
CO 2	0	1	0	0	0	0	2	0	0	0
CO 3	0	0	2	3	0	0	0	0	0	0
CO 4	2	0	0	0	0	0	0	0	0	0
CO 5	3	0	0	0	0	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT

Module	Units		Hrs	CO No.
1	1	**Communication Skills for Fitness Professionals**		
	1.1	Introduction to Soft Skills in Fitness Careers, Meaning, types, and importance of soft skills, Role in client satisfaction and retention	6	1
	1.2	Verbal and Non-Verbal Communication, Tone, clarity, and active listening, Body language, gestures, and facial expressions	6	2
	1.3	Presentation Skills for Trainers, Speaking to groups and leading classes, Confidence building in public communication	5	2
2	2	Interpersonal Skills and Client Management		
	2.1	Building Client Relationships, Trust, empathy, and rapport building, Understanding client needs and expectations	5	3
	2.2	Conflict Resolution and Problem-Solving, Handling complaints professionally, Negotiation skills for fitness professionals	5	3
	2.3	Time Management and Work Ethics, Prioritizing tasks, Maintaining punctuality and discipline	5	5
3	3	Leadership, Teamwork, and Ethics		
	3.1	Leadership in Fitness Environments, Leading group classes and teams, Motivating peers and clients	5	4
	3.2	Teamwork and Collaboration, Working with colleagues in health clubs, Cross-functional cooperation in fitness facilities	4	4
	3.3	Ethics, Inclusiveness, and Cultural Sensitivity, Professional ethics and boundaries, Serving diverse populations with equity and respect	4	5
4		**Teacher specific content**		

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** • **Lecture** • **Presentations** • **Interactive sessions** • **Class discussions**	
	MODE OF ASSESSMENT **Mode of Assessment**	
Assessment Type	**A. Continuous Comprehensive Assessment (CCA)** **CCA for Theory: 25 Marks** 1. **Written tests** 2. **Assignment** 3. **Seminar/ viva** **B. B. End Semester Evaluation (ESE)** **ESE for Theory: 50 Marks (1.5 Hrs)** **Written Test (50 Marks)** **Part A: Choose the Correct Answer (Answer all) - (10*1=10 Marks)** **Part B: Very Short Answer Questions (10 out of 12 Questions) - (10*2=20 Marks)** **Part C: Short Answer Questions (4 out of 6 Questions) - (4*5=20 Marks),**	

References

1. Bedi, M. (2019). Communication skills for professionals. Friends Publications India.
2. Kaur, R. (2020). Soft skills and personality development. Khel Sahitya Kendra.
3. Sharma, V. M. (2018). Interpersonal skills in sports and fitness. Sports Publication.
4. Singh, A. (2017). Professional ethics in physical education and sports. Friends Publications India.
5. Kumar, S. (2016). Leadership and team building in sports. Khel Sahitya Kendra.
6. Dhall, M. (2019). Client management in fitness industry. Sports Publication.

Suggested Readings

1. Bhardwaj, A. (2021). Communication and soft skills for physical educators. Friends Publications India.
2. Choudhary, R. (2017). Work ethics and professional conduct in sports. Khel Sahitya Kendra.
3. Jain, A. (2020). Time management for sports professionals. Sports Publication.
4. Sharma, A. (2018). Presentation skills and public speaking. Friends Publications India.

Mahatma Gandhi University Kottayam

Faculty/discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**INTERNSHIP**
Type of Course	**INT**
Course Code	**MG4INTMFT200**
Course Level	200
Course Summary	The internship is designed to provide students with real-world exposure and hands-on experience in professional environments aligned with their skill domain and major area of study. It acts as a vital link between academic learning and industry application, allowing students to apply theoretical concepts to practical situations. Through active engagement in industry, research institutions, or academic labs, students gain insights into organizational operations, workplace practices, and professional expectations. The internship also supports the development of key professional competencies such as communication, teamwork, time management, and ethical responsibility. Additionally, it encourages critical thinking, reflection, and self-assessment, helping students identify personal strengths and explore potential career pathways. Students shall undergo the internship in a Firm, Industry, or Organization, or engage in Training in Labs with faculty and researchers, or other Higher Education or Research Institutions, ensuring alignment with their area of academic specialization.

Semester	**4**	**Duration**	**60 hours**	**Credits**	**2**

COURSE OUTCOMES (CO)

CO No:	Expected Course Outcome	Learning Domains	PO No:
1	Demonstrate practical understanding of operational aspects in Their domain by engaging in real-world industry settings.	K	1,3,6,10
2	Apply academic knowledge and skills to identify and solve industry-relevant problems.	A	1,2,3,10
3	Exhibit professional competencies 106 ding effective communication, teamwork, time management, and ethical Responsibility.	S	4,5,8,9
4	Develop an understanding of workplace practices, expectations, and challenges.	U	1,6,10

5	Reflect critically on their internship experience to identify Personal strengths, growth areas, and career aspirations.	E	1,6,10

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**3**	**0**	**1**	**0**	**0**	**2**	**0**	**0**	**0**	**2**
CO 2	**3**	**2**	**3**	**0**	**0**	**0**	**0**	**0**	**0**	**2**
CO 3	**0**	**0**	**0**	**3**	**2**	**0**	**0**	**2**	**2**	**0**
CO 4	**3**	**0**	**0**	**0**	**0**	**2**	**0**	**0**	**0**	**2**
CO 5	**3**	**0**	**0**	**0**	**0**	**2**	**0**	**0**	**0**	**2**

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

Assessment Types	**A**	**Continuous Comprehensive Assessment(CCA)**	
		Components	**Marks**
		Feedback from the hosting organization	5
		Internal Supervisor feedback	10
		Total	**15**
	B	**END SEMESTER EVALUATION (ESE)**	
		Components	**Marks**
		Presentation	10
		Report	10
		Viva Voce	15
		Total	**35**

SEMESTER V

Semester 5

<table>
<tr><th rowspan="2">Course Code</th><th rowspan="2">Title of the Course</th><th rowspan="2">Type of the Course</th><th rowspan="2">Credit</th><th rowspan="2">Hours /week</th><th colspan="3">Hour Distribution /week</th></tr>
<tr><th>L</th><th>P</th><th>O</th></tr>
<tr><td>MG5SDCMFT300</td><td>Fitness facility & Client Management</td><td>SDC</td><td>4</td><td>4</td><td>4</td><td>0</td><td>0</td></tr>
<tr><td>MG5SDEMFT300</td><td>Group Fitness & Mind-Body Practices</td><td rowspan="2">SDE (Elective papers-selection one)</td><td rowspan="2">4</td><td rowspan="2">5</td><td rowspan="2">3</td><td rowspan="2">2</td><td rowspan="2">0</td></tr>
<tr><td>MG5SDEMFT301</td><td>Massage Manipulation: Theory and Practice</td></tr>
<tr><td>MG5SDCMFT301</td><td>Foundation of Core Strengthening & Pilates</td><td>SDC</td><td>4</td><td>5</td><td>3</td><td>2</td><td>0</td></tr>
<tr><td>MG5SECMFT300</td><td>Mastering Sports Skill and Techniques</td><td>SEC</td><td>3</td><td>4</td><td>2</td><td>2</td><td>0</td></tr>
<tr><td>MG5MPCMFT300</td><td>Scientific Principles of Fitness</td><td>MPC</td><td>4</td><td>4</td><td>4</td><td>0</td><td>0</td></tr>
<tr><td>MG5VACMFT300</td><td>Exercise Science for Women and Children</td><td>VAC</td><td>3</td><td>3</td><td>3</td><td>0</td><td>0</td></tr>
</table>

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Mahatma Gandhi University Kottayam

<table>
<tr><td>Faculty/ Discipline</td><td colspan="5">Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training</td></tr>
<tr><td>Programme</td><td colspan="5">B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training</td></tr>
<tr><td>Course Name</td><td colspan="5">Fitness Facility & Client Management</td></tr>
<tr><td>Type of Course</td><td colspan="5">SDC</td></tr>
<tr><td>Course Code</td><td colspan="5">MG5SDCMFT300</td></tr>
<tr><td>Course Level</td><td colspan="5">300</td></tr>
<tr><td>Course Summary</td><td colspan="5">This course provides a foundational understanding of how to design, operate, and manage a fitness facility, along with skills required to handle client interactions effectively. It includes the study of layout planning, types of equipment (cardio, resistance, free weights, and accessories), scheduling, maintenance, and basic client management skills. Students will learn how to create a professional training environment and deliver customer satisfaction through appropriate communication, facility safety, and personalized service.</td></tr>
<tr><td>Semester</td><td>5</td><td colspan="2">Credits</td><td>4</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course Details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>4</td><td>0</td><td>0</td><td>60</td></tr>
<tr><td>Pre-requisites, if any</td><td colspan="5">Students must have completed basic courses in Principles of Physical Fitness, Exercise Programming, and Introduction to Fitness Training.</td></tr>
</table>

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	**Identify various types of fitness facilities and their essential components.**	**K**	1
CO2	Understand and categorize gym/studio equipment, cardio machines, resistance machines, and accessories.	**K**	**4,5**
CO3	Design and organize functional training spaces based on client needs and safety.	**K**	**2,3**
CO4	Demonstrate knowledge of facility operations, equipment maintenance, and scheduling.	**U**	**3,5,6**

CO5	Apply effective communication and client relationship management strategies in a fitness setting.	**U**	**2,4**
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**
CO1	2	0	0	0	0	0	0	0	0	0
CO2	0	0	0	1	3	0	0	0	0	0
CO3	0	2	1	0	0	0	0	0	0	0
CO4	0	0	**2**	0	2	2	0	0	0	0
CO5	0	2	0	2	0	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Introduction to Fitness Facility Planning**		
	1.1	Overview of Fitness Industry & Facility Types	3	1
	1.2	Space Allocation: Cardio, Strength, Group Training Zones	3	2
	1.3	Basic Layout Planning and Interior Design	3	2
	1.4	Ventilation, Lighting, Flooring & Sound Systems	4	2
	1.5	Legal Considerations: Insurance, Licenses, Safety Policies	4	2
2	2	**Fitness Equipment and Functional Areas**		
	2.1	Cardio Equipment: Treadmills, Bikes, Rowers, Stair Climbers	3	1
	2.2	Resistance Equipment: Selectorized Machines, Plate-loaded Systems	3	**1**
	2.3	Free Weights: Dumbbells, Barbells, Racks	4	**4**
	2.4	Accessories: Kettlebells, Medicine Balls, Bands, BOSU	4	**3**

	2.5	Functional Fitness Zones: Plyo Boxes, Battle Ropes, TRX Systems	3	**5**
3	3	**Facility Operation and Maintenance**		
	3.1	Equipment Setup, Calibration, and Maintenance Schedules	3	1
	3.2	Daily Cleaning and Hygiene Protocols	3	1
	3.3	Safety Inspections and Risk Management	2	5
	3.4	Staff Roles: Front Desk, Trainers, Floor Managers	3	3
	3.5	Technology in Facility Management: Software & Apps	3	4
4	4	**Client Management and Engagement**		
	4.1	Types of Clients: Beginners, Seniors, Special Populations	2	3
	4.2	Fitness Orientation and Onboarding Processes	2	4
	4.3	Record Keeping: Client Profiles, Progress Logs	2	2
	4.4	Goal Setting and Motivation Techniques	3	3
	4.5	Feedback, Complaint Handling, and Service Recovery	3	2
5	**TEACHER SPECIFIC CONTENT**			

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● Lecture (Power Point presentation) ● Demonstration ● Peer teaching ● Lesson plans ● Group Activities
	Mode of Assessment
	MODE OF ASSESSMENT Continuous Comprehensive Assessment (CCA) **CCA for Theory:30 Marks** CCA -15 marks (Assignment, viva) CCA -15 mark, (Presentation, individual involvement)
Assessment Type	**End Semester Examination (ESE)** **ESE Theory –70 marks (Written examination theory)** **Duration of Examination 2 hrs** **Different part of examinations** – **Part 1**, MCQ 10x1=10 (number of questions 12-number of questions to be answered 10) **Part 2** - Short Answer – marks - 5x2=10 (number of questions 7 -number of questions to be answered 5) **Part 3 -** Short Essay - 4x5=20 (number of questions 6-number of questions to be answered 4) **Part 4** - Essay-2 x 15=30 - (number of questions 3-number of questions to be answered 2).

References

1. Baechle, T. R., & Earle, R. W. (2022). Essentials of strength training and conditioning (5th ed., pp. 220–240). Human Kinetics.
2. Chapman, J. (2022). Client-centered exercise prescription (3rd ed., pp. 130–150). Human Kinetics.
3. Forsyth, E. (2020). Fitness management: A comprehensive resource for managing and operating programs and facilities (4th ed., pp. 180–200). Human Kinetics.
4. Green, M. (2023). Business and management in fitness facilities (pp. 100–120). Routledge.
5. Howley, E. T., & Thompson, D. L. (2022). Fitness professional’s handbook (8th ed., pp. 290–310). Human Kinetics.

Suggested Readings

1. Kravitz, L. (2021). Programming and instruction for personal trainers (2nd ed., pp. 165–185). Human Kinetics.
2. Melton, D. I., & Katula, J. A. (2020). Fitness facility management (2nd ed., pp. 120–140). Sagamore Publishing.
3. Ratamess, N. A. (2021). ACSM's foundations of strength training and conditioning (2nd ed., pp. 200–220). Wolters Kluwer.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training				
Programme	B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training				
Course Name	Group Fitness & Mind-Body Practices				
Type of Course	SDE				
Course Code	MG5SDEMFT300				
Course Level	300				
Course Summary	This course introduces students to the principles, methods, and benefits of Group Fitness Training and various Mind-Body practices such as yoga, pilates, tai chi, and meditation. The course emphasizes both theoretical foundations and practical skills, aiming to foster physical fitness, mental well-being, and community participation through group-based fitness routines. Students will develop competency in leading group fitness classes, integrating mind-body techniques, and adapting programs for diverse populations.				
Semester	5	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	75
Pre-requisites, if any	Basic knowledge of human anatomy, exercise science, and fitness training principles covered in earlier semesters.				

COURSE OUTCOMES (CO)			
CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	**Understand the science and structure behind group fitness.**	**K**	2,3
CO2	Acquire skills to plan, lead, and evaluate group sessions.	**S**	**3,4**
CO3	Learn to integrate mind-body techniques to enhance overall well-being.	**U**	**1,2**
CO4	Explore stress management through movement and mindfulness.	**U**	**1,10**
CO5	Build readiness for employment in the fitness and wellness industry.	**K**	**5**
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	0	2	1	0	0	0	0	0	0	0
CO2	**0**	0	2	1	0	0	0	0	0	0
CO3	1	2	0	0	0	0	0	0	0	0
CO4	1	0	0	0	0	0	0	0	0	2
CO5	0	0	0	0	3	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Introduction to Group Fitness and Mind-Body Practices**		
	1.1	History and Evolution of Group Fitness	4	2
	1.2	Philosophy and Principles of Mind-Body Practices	4	3
	1.3	Types of Group Fitness Formats (Aerobics, Step, Dance, HIIT)	5	1
	1.4	Health Benefits of Group Fitness and Mind-Body Training	5	2
	1.5	Roles and Responsibilities of a Group Fitness Instructor	4	4
2	2	**Planning and Instructional Design for Group Fitness**		
	2.1	Components of a Group Fitness Session (Warm-up, Main, Cooldown)	4	3
	2.2	Music Selection and Cueing Techniques	4	3
	2.3	Safety Guidelines and Risk Management	3	4
	2.4	Progression and Regression Strategies for Various Populations	4	2
	2.5	Class Formats: Circuit, Bootcamp, Functional Training, etc.	4	2
3	3	**Mind-Body Movement Practices**		
	3.1	Yoga: Asanas, Pranayama, and Relaxation Techniques	4	3
	3.2	Pilates: Core Engagement, Posture, and Breathing	4	5
	3.3	Tai Chi: Flow, Movement Meditation, and Balance	4	2
	3.4	Meditation and Mindfulness Techniques	5	2
	3.5	Integration of Mind-Body Practices in Group Classes	5	2
4	4	**Anatomy, Physiology, and Psychology of Mind-Body Fitness**		
	4.1	Musculoskeletal System and Movement Patterns	2	1
	4.2	Respiratory and Cardiovascular Benefits	2	2
	4.3	Neuropsychology of Exercise and Mindfulness	2	3
	4.4	Impact on Stress, Anxiety, and Emotional Well-being	3	2
	4.5	Sleep, Recovery, and Hormonal Balance through Mind-Body Fitness	3	5
5		**TEACHER SPECIFIC CONTENT**		

<table>
<tr><td rowspan="2">Teaching and Learning

Approach</td><td>Classroom Procedure (Mode of transaction)

Mode of Assessment</td></tr>
<tr><td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
● Lesson plans
● Hands on training</td></tr>
<tr><td rowspan="2">Mode of Assessment</td><td>Continues Comprehensive Assessment (CCA)
Total Mark - 30

<table><tr><th>Assessment Method (CCA)</th><th>Marks</th></tr><tr><td>Assignment</td><td>15</td></tr><tr><td>Individual Involvement & Viva</td><td>15</td></tr><tr><td>Total</td><td>30</td></tr></table></td></tr>
<tr><td>End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 70
Duration – 3 hrs

<table><tr><th>Assessment Method (ESE)</th><th>Marks</th></tr><tr><td>Lesson Plan</td><td>30</td></tr><tr><td>Demonstration & Presentation</td><td>30</td></tr><tr><td>Viva</td><td>10</td></tr><tr><td>Total</td><td>70</td></tr></table></td></tr>
</table>

References

1. American College of Sports Medicine. (2018). ACSM's group exercise instructor manual: A guide for fitness professionals (5th ed., pp. 140–160). Wolters Kluwer.
2. Brown, C. (2003). The yoga bible: The definitive guide to yoga postures (pp. 210–230). Walking Stick Press.
3. Feuerstein, G. (2003). The deeper dimension of yoga: Theory and practice (pp. 120–140). Shambhala Publications.
4. Fishman, L. M., & Saltonstall, M. (2008). Yoga for osteoporosis: The complete guide (pp. 90–110). Turner Publishing Company.
5. Howley, E. T., & Thompson, D. L. (2021). Fitness professional's handbook (8th ed., pp. 245–265). Human Kinetics.

Suggested Readings

1. Robinson, L. (2013). The Pilates bible: The most comprehensive and accessible guide to Pilates ever (pp. 170–190). Octopus Publishing.
2. Stolberg, R. L. (2021). Group fitness instructor training manual (pp. 115–135). Kendall Hunt Publishing.
3. Wayne, P. M., & Fuerst, M. L. (2013). The Harvard Medical School guide to Tai Chi: 12 weeks to a healthy body, strong heart, and sharp mind (pp. 200–220). Shambhala Publications.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc .(Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Massage Manipulation: Theory and Practice**
Type of Course	**SDE**
Course Code	**MG5SDEMFT301**
Course Level	300
Course Summary	This course provides theoretical knowledge and practical skills in massage manipulation for sports, therapeutic, and wellness contexts. Students will learn anatomy-relevant massage applications, manipulation techniques, contraindications, and client care protocols. The course integrates evidence-based principles with hands-on training to prepare students for professional practice in sports and fitness environments.

Semester	5	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	Basic knowledge of human anatomy, exercise science, and fitness training principles covered in earlier semesters.				

COURSE OUTCOMES (CO)			
CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	Explain the scope, principles, and ethics of massage manipulation in sports and wellness.	**U**	2,3
CO2	Demonstrate knowledge of anatomy and physiology relevant to massage techniques.	**An**	**3,4**
CO3	Apply various massage manipulation techniques for sports performance and recovery.	**A**	**1,2**
CO4	Assess client needs, plan, and deliver appropriate massage interventions.	**An**	**1,10**
CO5	Evaluate massage outcomes and adapt techniques for safety, effectiveness, and client satisfaction.	**K**	**5**
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO -PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	0	2	2	0	0	0	0	0	0	0
CO2	**0**	0	2	1	0	0	0	0	0	0
CO3	2	1	0	0	0	0	0	0	0	0
CO4	2	0	0	0	0	0	0	0	0	3
CO5	0	0	0	0	2	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	Introduction to Massage Manipulation		
	1.1	History and evolution of massage therapy	4	1
	1.2	Scope and applications in sports and fitness	4	3
	1.3	Principles and ethics of massage manipulation	4	2
	1.4	Role of massage in injury prevention and recovery	5	2
	1.5	Legal considerations and professional conduct	5	1
2	2	Anatomy and Physiology for Massage		
	2.1	Musculoskeletal system overview	4	3
	2.2	Circulatory and lymphatic system functions in massage	4	2
	2.3	Nervous system response to touch and pressure	5	4
	2.4	Soft tissue types and properties	5	2
	2.5	Functional anatomy relevant to sports massage	4	4
3	3	Massage Manipulation Techniques		
	3.1	Effleurage (stroking) methods and purposes	4	3
	3.2	Petrissage (kneading) techniques	4	5
	3.3	Friction and vibration techniques	3	4
	3.4	Tapotement (percussion) variations	4	2
	3.5	Myofascial release and trigger point therapy basics	4	2
4	4	Practical Applications in Sports and Fitness		
	4.1	Pre-event sports massage protocols	2	5
	4.2	Post-event recovery massage	2	2
	4.3	Rehabilitation massage techniques	2	3
	4.4	Massage for flexibility and range of motion	3	4
	4.5	Adapting massage for different sports disciplines	3	5
5		**TEACHER SPECIFIC CONTENT**		

<table>
<tr><td rowspan="3">Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)</td></tr>
<tr><td>Mode of Assessment</td></tr>
<tr><td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
● Lesson plans
● Hands on training</td></tr>
<tr><td rowspan="2">Mode of Assessment</td><td>Continues Comprehensive Assessment (CCA)
Total Mark - 30
<table>
<tr><th>Assessment Method (CCA)</th><th>Marks</th></tr>
<tr><td>Assignment</td><td>15</td></tr>
<tr><td>Individual Involvement & Viva</td><td>15</td></tr>
<tr><td>Total</td><td>30</td></tr>
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<tr><td>End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 70
Duration – 3 hrs
<table>
<tr><th>Assessment Method (ESE)</th><th>Marks</th></tr>
<tr><td>Lesson Plan</td><td>30</td></tr>
<tr><td>Demonstration & Presentation</td><td>30</td></tr>
<tr><td>Viva</td><td>10</td></tr>
<tr><td>Total</td><td>70</td></tr>
</table></td></tr>
</table>

References

1. Cushion, C., & Lyle, J. (2023). Sports coaching concepts: A framework for coaching practice (3rd ed., pp. 150–170). Routledge.
2. Haywood, K. M., & Getchell, N. (2021). Life span motor development (7th ed., pp. 200–220). Human Kinetics.
3. Jones, R. L., & Kingston, K. (2022). Developing skill in sport: An introduction (2nd ed., pp. 115–135). Routledge.
4. McGinnis, P. M. (2023). Biomechanics of sport and exercise (3rd ed., pp. 180–200). Human Kinetics.
5. Nash, C. (2020). Practical sports coaching (pp. 90–110). Routledge.

Suggested Readings

1. Sheel, A. W. (2022). Fundamentals of physical activity and sport science (pp. 140–160). Springer.
2. Vickery, W. (2022). Recreational sports programming (pp. 125–145). Human Kinetics.
3. Weese, W. J., & Jamieson, L. M. (2022). Introduction to sport and leisure management (pp. 170–190). Human Kinetics.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Foundation of Core Strengthening & Pilates**
Type of Course	**SDC**
Course Code	**MG5SDCMFT301**
Course Level	300
Course Summary	This course introduces students to the fundamentals of core strengthening and Pilates-based training techniques. The emphasis is placed on understanding the core musculature, principles of posture, breathing control, and stabilization. Through both theoretical lectures and practical sessions, students will learn structured exercise routines, modern Pilates applications, and functional core conditioning for performance, rehabilitation, and overall fitness.

Semester	5	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	Basic understanding of human anatomy, posture, and introductory physical fitness training.				

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	**Identify and describe the anatomical structure and functional role of core muscles in fitness.**	**U**	1,2,10
CO2	Demonstrate knowledge of Pilates principles and their relevance to physical fitness.	**U**	**1,2**
CO3	Design and perform progressive core strengthening exercises for different fitness levels	**U**	**2,4,10**
CO4	Apply breathing, control, and alignment principles in Pilates for functional outcomes.	**K**	**10**
CO5	Integrate core training into fitness programming for strength, posture, and injury prevention.	**E**	**1,10**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	1	2	0	0	0	0	0	0	0	2
CO2	**2**	2	0	0	0	0	0	0	0	0
CO3	0	2	0	2	0	0	0	0	0	3
CO4	0	0	0	0	0	0	0	0	0	1
CO5	2	0	0	0	0	0	0	0	0	3
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Core Anatomy and Functional Relevance**		
	1.1	Introduction to Core and Postural Muscles	4	2
	1.2	Muscles of Stabilization and Movement	4	2
	1.3	Importance of Core Strength in Daily Life and Sports	5	1
	1.4	Functional Anatomy of Lower Back, Pelvis, and Abdominals	5	3
	1.5	Effects of Weak Core Muscles and Common Postural Issues	4	3
2	2	**Principles and Fundamentals of Pilates**		
	2.1	History and Evolution of Pilates	4	1
	2.2	Six Principles of Pilates: Centering, Control, Precision, Breath, Flow, Concentration	4	2
	2.3	Breathing and Mind-Body Connection	3	5
	2.4	Introduction to Mat-Based Pilates	4	3
	2.5	Pilates Warm-up and Safety Guidelines	4	4
3	3	**Core Conditioning Techniques and Training Protocols**		
	3.1	Static and Dynamic Core Exercises (Plank, Bridge, Bird Dog, etc.)	4	5

	3.2	Core Engagement and Activation Drills	4	2
	3.3	Core Stability vs. Core Strength Training	4	1
	3.4	Progressions and Regressions for Core Training	5	2
	3.5	Circuit Training and Functional Core Workouts	5	3
4	4	**Pilates Exercise Practice and Programming**		
	4.1	Mat-Based Pilates: Beginner to Intermediate Sequences	2	1
	4.2	Use of Small Apparatus: Pilates Rings, Resistance Bands, Foam Rollers	2	2
	4.3	Corrective Exercises for Postural Alignment	2	5
	4.4	Pilates for Special Populations (Back Pain, Seniors, Athletes)	3	3
	4.5	Designing a Weekly Pilates Program	3	4
5		**TEACHER SPECIFIC CONTENT**		

<table>
<tr><td rowspan="3">Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)</td></tr>
<tr><td>Mode of Assessment</td></tr>
<tr><td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
● Lesson plans
● Group Activities</td></tr>
<tr><td>Mode of Assessment</td><td>Continues Comprehensive Assessment (CCA)
Total Mark - 30
<table><tr><th>Assessment Method (CCA)</th><th>Marks</th></tr><tr><td>Assignment</td><td>15</td></tr><tr><td>Individual Involvement & Viva</td><td>15</td></tr><tr><td>Total</td><td>30</td></tr></table></td></tr>
</table>

End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 70
Duration – 3 hrs

Assessment Method (ESE)	**Marks**
Lesson Plan	30
Demonstration & Presentation	30
Viva	10
Total	70

References

1. Barker, K. L., & Ersser, S. J. (2022). Functional training and core stability (3rd ed., pp. 145–165). Elsevier Health Sciences.
2. Freeman, J. (2023). Core training for sport and performance (pp. 110–130). Routledge.
3. Isacowitz, R. (2022). Pilates anatomy (2nd ed., pp. 90–110). Human Kinetics.
4. Kolber, M. J., & Beekhuizen, K. S. (2020). Core assessment and training (2nd ed., pp. 200–220). Human Kinetics.
5. Latey, P. (2021). The science of Pilates: Understand the techniques and philosophy of Pilates (pp. 95–115). DK Publishing.

Suggested Readings

1. Muscolino, J. E. (2021). The muscular system manual: The skeletal muscles of the human body (5th ed., pp. 310–330). Elsevier.
2. Siler, B. (2021). The Pilates body: The ultimate at-home guide to strengthening, lengthening, and toning your body (pp. 140–160). Broadway Books.
3. Stott, M. (2022). Essential Pilates: Your guide to strength, flexibility, and posture (pp. 80–100). DK Publishing.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences**
Programme	**B.Voc .(Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Mastering Sports Skill and Techniques**
Type of Course	**SEC**
Course Code	**MG5SECMFT300**
Course Level	300
Course Summary	This course focuses on helping students master the fundamental techniques and skills in major and leisure games, with an emphasis on practical learning and enjoyment. It introduces students to basic gameplay and movement skills in popular sports like Football, Basketball, Volleyball, Kho-Kho, Badminton, Table Tennis, and Cricket, without requiring a deep understanding of rules and officiating. Through structured drills, cooperative games, and recreational sports, students learn how physical fitness is enhanced through play, skill repetition, and sport-specific movement.

<table>
<tr><td>Semester</td><td>5</td><td colspan="2">Credits</td><td>3</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course Details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>2</td><td>1</td><td>0</td><td>60</td></tr>
<tr><td>Pre-requisites, if any</td><td colspan="5">Basic motor coordination and physical fitness, Interest in games and physical activity, Willingness to participate in team and individual sports, No competitive experience required</td></tr>
</table>

COURSE OUTCOMES (CO)

CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	**Demonstrate foundational skills and movement patterns in major indoor and outdoor sports.**	**U**	4,5
CO2	Use sport-specific techniques to improve coordination, agility, and cardiovascular fitness.	**C**	**2,3,4**
CO3	Participate actively in recreational and leisure games for general well-being.	**U**	**3,4**
CO4	Apply learned techniques in a safe and cooperative manner in individual and team activities.	**U**	**1,2**
CO5	Develop lifelong fitness habits through engaging and enjoyable gameplay experiences.	**U**	**2,5**
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	0	0	0	**1**	3	0	0	0	0	0
CO2	0	2	2	1	0	0	0	0	0	0
CO3	0	**0**	1	2	**0**	0	0	0	0	0
CO4	1	2	0	0	0	0	0	0	0	0
CO5	0	1	0	0	3	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Introduction to Skill-Based Play and Leisure Fitness**		
	1.1	Role of games in holistic fitness	5	1
	1.2	Concepts of leisure games and active recreation	5	1
	1.3	Benefits of skill-based physical activities	5	1
	1.4	Warm-up and mobility drills for skill preparation	4	2
	1.5	Cool-down and flexibility for recovery	4	1
2	2	**Movement & Skill Techniques in Major Team Sport & Games**		
	2.1	Football – Passing, dribbling, ball control drills	4	2
	2.2	Basketball – Ball handling, passing, shooting drills	4	1
	2.3	Volleyball – Underarm pass, overhand pass, basic spike	5	1
	2.4	Cricket – Batting stance, bowling grip, catching drills, fielding drills	2	1
	2.5	Athletics - Walking, Running Partner and small-group activities	2	1
3	3	**Fundamentals of Indigenous and Court-Based Games**		
	3.1	Kho-Kho – Running, sitting, turning, chasing	4	1

	3.2	Badminton – Grip, basic strokes, footwork drills	4	1
	3.3	Table Tennis – Grip, forehand/backhand basics	4	2
	3.4	Coordination and reflex improvement through racquet games	4	2
	3.5	Fun relay and shuttle-based games	4	2
4		**TEACHER SPECIFIC CONTENT**		

<table>
<tr><td rowspan="3">Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)</td></tr>
<tr><td>Mode of Assessment</td></tr>
<tr><td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
● Lesson plans
● Group Activities</td></tr>
<tr><td rowspan="2">Mode of Assessment</td><td>Continues Comprehensive Assessment (CCA)
Total Mark - 25 marks
<table><tr><th>Assessment Method (CCA)</th><th>Marks</th></tr><tr><td>Assignment</td><td>10</td></tr><tr><td>Individual Involvement & Viva</td><td>15</td></tr><tr><td>Total</td><td>25</td></tr></table></td></tr>
<tr><td>End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 50
Duration – 2 hrs
<table><tr><th>Assessment Method (ESE)</th><th>Marks</th></tr><tr><td>Lesson Plan</td><td>20</td></tr><tr><td>Demonstration & Presentation</td><td>20</td></tr><tr><td>Viva</td><td>10</td></tr><tr><td>Total</td><td>50</td></tr></table></td></tr>
</table>

References

1. Cushion, C., & Lyle, J. (2023). Sports coaching concepts: A framework for coaching practice (3rd ed., pp. 150–170). Routledge.
2. Haywood, K. M., & Getchell, N. (2021). Life span motor development (7th ed., pp. 200–220). Human Kinetics.
3. Jones, R. L., & Kingston, K. (2022). Developing skill in sport: An introduction (2nd ed., pp. 115–135). Routledge.
4. McGinnis, P. M. (2023). Biomechanics of sport and exercise (3rd ed., pp. 180–200). Human Kinetics.

Suggested Readings

1. Cushion, C., & Lyle, J. (2023). Sports coaching concepts: A framework for coaching practice (3rd ed., pp. 150–170). Routledge.
2. Haywood, K. M., & Getchell, N. (2021). Life span motor development (7th ed., pp. 200–220). Human Kinetics.
3. Jones, R. L., & Kingston, K. (2022). Developing skill in sport: An introduction (2nd ed., pp. 115–135). Routledge.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Scientific Principles of Fitness**
Type of Course	**MPC**
Course Code	**MG5MPCMFT300**
Course Level	300
Course Summary	This course provides students with a scientific understanding of physical fitness, covering the foundational principles that govern exercise, training, and body function. It introduces basic anatomy, physiology, kinesiology, and touches upon elements of gym training, yoga, and movement science. Through this course, students will learn how the body responds to different forms of exercise and how to apply scientific principles for safe and effective fitness training.

Semester	5	**Credits**		4	**Total Hours**
Course details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		4	0	0	**60**
Pre-requisites, if any	Students should have basic knowledge of physical activity, and an interest in fitness, health, or sports. No prior expertise in anatomy or exercise science is required.				

COURSE OUTCOMES (CO)

CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	**Understand the fundamental principles and components of physical fitness.**	U	2
CO2	Explain how the body systems (muscular, skeletal, cardiovascular) function during exercise.	U	2
CO3	Analyze movement and apply basic kinesiology in fitness programs.	A	2
CO4	Identify the scientific basis behind gym exercises, yoga, and fitness plans	A	2
CO5	Design simple fitness programs based on training principles and body responses.	U	2

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	0	2	0	0	0	0	0	0	0	0
CO2	0	1	0	0	0	0	0	0	0	0
CO3	0	**2**	0	0	0	0	0	0	0	0
CO4	0	1	0	0	0	0	0	0	0	0
CO5	0	3	0	0	0	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Introduction to Fitness and Health**		
	1.1	Definition and components of physical fitness	3	2
	1.2	Physical activity, exercise, and wellness	3	3
	1.3	Health-related vs skill-related fitness	3	2
	1.4	Benefits of regular exercise	4	1
	1.5	Basic lifestyle habits for lifelong fitness	4	1
2	2	**Basic Anatomy and Physiology in Fitness**		
	2.1	Muscular system: types and roles in movement	3	2
	2.2	Skeletal system: joints, bones, and posture	3	3
	2.3	Cardiovascular and respiratory response to exercise	4	4
	2.4	Energy systems and metabolism	4	1
	2.5	Introduction to body composition and body types	3	2
3	3	**Principles of Exercise Training**		
	3.1	Warm-up, cool-down, and recovery	3	2
	3.2	FITT principle (Frequency, Intensity, Time, Type)	3	3
	3.3	Principles of overload, specificity, and progression	2	2
	3.4	Periodization and training plans	3	1
	3.5	Safety, hydration, and injury prevention	3	2
4	4	**Kinesiology and Biomechanics in Fitness**		
	4.1	Levers in human movement	2	1
	4.2	Basic movement planes and joint actions	2	2
	4.3	Posture and alignment in exercise	2	1
	4.4	Common movement dysfunctions and corrections	3	2
	4.5	Application in gym workouts and yoga	3	3
5		**TEACHER SPECIFIC CONTENT**		

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** ● **Lecture (Power Point presentation)** ● **Demonstration** ● **Peer teaching** ● **Lesson plans** ● **Hands on training**
	Mode of Assessment
	MODE OF ASSESSMENT Continuous Comprehensive Assessment (CCA) **CCA for Theory:30 Marks** CCA -15 marks (Assgnment, viva) CCA -15 mark, (Presentation, individual involvement)
Assessment Type	**End Semester Examination (ESE)** **ESE Theory –70 marks (Written examination theory)** **Duration of Examination 2 hrs** **Different part of examinations** – **Part 1**, MCQ 10x1=10 (number of questions 12-number of questions to be answered 10) **Part 2** - Short Answer – marks - 5x2=10 (number of questions 7 -number of questions to be answered 5) **Part 3 -** Short Essay - 4x5=20 (number of questions 6-number of questions to be answered 4) **Part 4** - Essay-2 x 15=30 - (number of questions 3-number of questions to be answered 2).

References

1. Abernathy, P. (2023). Principles of exercise science and fitness training (pp. 145–160). Routledge.
2. Floyd, R. T. (2022). Manual of structural kinesiology (21st ed., pp. 100–115). McGraw-Hill Education.
3. Hall, S. J. (2022). Basic biomechanics (9th ed., pp. 220–240). McGraw-Hill Education.
4. Haff, G. G., & Triplett, N. T. (2022). Essentials of strength training and conditioning (4th ed., pp. 305–325). Human Kinetics.
5. Kenney, W. L., Wilmore, J. H., & Costill, D. L. (2021). Physiology of sport and exercise (7th ed., pp. 180–195). Human Kinetics.

Suggested Readings

6. McArdle, W. D., Katch, F. I., & Katch, V. L. (2022). Exercise physiology: Nutrition, energy, and human performance (9th ed., pp. 510–525). Wolters Kluwer.
7. Saraswati, S. S. (2021). Asana pranayama mudra bandha (4th ed., pp. 230–250). Yoga Publications Trust.
8. Thompson, W. R. (2023). ACSM’s guidelines for exercise testing and prescription (11th ed., pp. 115–130). Wolters Kluwer.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc .(Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Exercise Science for Women and Children**
Type of Course	**VAC**
Course Code	**MG5VACMFT300**
Course Level	300
Course Summary	This course provides specialized knowledge and hands-on skills for designing and delivering safe, effective exercise programs tailored for **women and children**. It focuses on the physiological, psychological, and developmental differences that impact training outcomes. Students will gain practical experience in creating fitness solutions across life stages—pregnancy, postnatal, prepubescence, and adolescence.

Semester	5	**Credits**		3	**Total Hours**
Course details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	0	0	**45**
Pre-requisites, if any	Students must have a basic understanding of **anatomy, physiology, and general fitness training principles** as covered in first-year vocational courses.				

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	Understand sex-specific and age-specific physiological responses to exercise	U	1,2,10
CO2	Assess and address the unique fitness needs of women (including pregnant and postpartum stages).	U	2,4,5
CO3	Design age-appropriate physical activity programs for children	AN	2,4
CO4	Apply safety considerations and injury-prevention strategies for both groups.	A	2,4,5
CO5	Demonstrate practical instruction techniques with communication suited to women and children.	U	2,4,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	1	2	0	0	0	0	0	0	0	2
CO2	**0**	2	0	1	2	0	0	0	0	0
CO3	0	**3**	0	2	0	0	0	0	0	0
CO4	0	2	0	3	2	0	0	0	0	0
CO5	0	1	0	2	0	0	0	0	0	2
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	Foundations of Exercise Science for Women & Children		
	1.1	Gender and Age Differences in Anatomy and Physiology	5	1
	1.2	Hormonal Influence on Physical Performance (e.g., Menstrual Cycle, Pregnancy)	4	3
	1.3	Growth and Motor Development in Children	4	2
	1.4	Psychological and Social Aspects of Exercise for Women and Children	4	3
2	2	Assessment and Screening Techniques		
	2.1	Health and Risk Screening for Pregnant and Postpartum Women	4	1
	2.2	Functional and Developmental Fitness Assessments in Children	4	3
	2.3	Body Composition, Flexibility, Strength and Cardio Assessments	4	4
	2.4	Interpreting Growth Charts and Maturation Stages	5	5
3	3	Exercise Programming for Women		
	3.1	General Fitness Programs for Adult Women	4	1
	3.2	Pre- and Postnatal Fitness Guidelines	3	1
	3.3	Strength and Mobility Training for Osteoporosis Prevention	2	2
	3.4	Group Fitness and Yoga for Women	2	3
4		**TEACHER SPECIFIC CONTENT**		

<table>
<tr><td rowspan="3">Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)</td></tr>
<tr><td>Mode of Assessment</td></tr>
<tr><td>Continues Comprehensive Assessment (CCA)
Total Mark - 25
<table><tr><th>Assessment Method (CCA)</th><th>Marks</th></tr><tr><td>Assignment</td><td>10</td></tr><tr><td>Individual Involvement & Viva</td><td>15</td></tr><tr><td>Total</td><td>25</td></tr></table></td></tr>
<tr><td>Assessment Type</td><td>End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 50
Duration – 2 hrs
<table><tr><th>Assessment Method (ESE)</th><th>Marks</th></tr><tr><td>Lesson Plan</td><td>20</td></tr><tr><td>Demonstration & Presentation</td><td>20</td></tr><tr><td>Viva</td><td>10</td></tr><tr><td>Total</td><td>50</td></tr></table></td></tr>
</table>

References

1. Faigenbaum, A. D., & Myer, G. D. (2022). *Youth resistance training: Strengthening young athletes*. Human Kinetics. **(pp. 45–95)**
2. ACSM. (2021). *ACSM's guidelines for exercise testing and prescription* (11th ed.). Wolters Kluwer. **(pp. 224–263)**
3. Kenney, W. L., Wilmore, J. H., & Costill, D. L. (2023). *Physiology of sport and exercise* (8th ed.). Human Kinetics. **(pp. 410–440)**
4. Artal, R., & O'Toole, M. (2021). *Exercise in pregnancy*. Springer. **(pp. 35–80)**
5. Howley, E. T., & Thompson, D. L. (2022). *Fitness professional's handbook* (8th ed.). Human Kinetics. **(pp. 352–386)**

Suggested Readings

1. Nield, M., & Straker, L. (2022). *Children, exercise and fitness*. Routledge. **(pp. 60–110)**
2. Reiff, G. G. (2021). *Exercise and the female life cycle*. Wolters Kluwer. **(pp. 88–132)**
3. Rowland, T. W. (2020). *Children's exercise physiology* (2nd ed.). Human Kinetics. **(pp. 105–160)**

SEMESTER VI

Semester 6

Course Code	Title of the Course	Type of the Course	Credit	Hours /week	Hour Distribution /week		
					L	P	O
MG6SDEMFT300	Mental Health and Physical Fitness Dynamics	SDE(Elective papers-selection one)	4	5	3	2	0
MG6SDEMFT301	Adventure sports and fitness						
MG6SDCMFT300	Advanced Training designs & Periodization	SDC	4	5	3	2	0
MG6SECMFT300	Officiating and Refereeing in Combat Sports (Judo/ Taekwondo/ Karate)	SEC	3	3	3	0	0
MG6MPCMFT300	Fitness Centre Management	MPC	4	4	4	0	0
MG6VACMFT300	Basics of Geriatric Fitness	VAC	3	3	3	0	0
MG6PRJMFT300	Project	PRJ	4	8	0	8	0

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

Mahatma Gandhi University Kottayam

<table>
<tr><td>Faculty/ Discipline</td><td colspan="5">Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training</td></tr>
<tr><td>Programme</td><td colspan="5">B.Voc . (Honours) Advanced Course in Multi Sports and Fitness Training</td></tr>
<tr><td>Course Name</td><td colspan="5">Mental Health and Physical Fitness Dynamics</td></tr>
<tr><td>Type of Course</td><td colspan="5">SDE</td></tr>
<tr><td>Course Code</td><td colspan="5">MG6SDEMFT300</td></tr>
<tr><td>Course Level</td><td colspan="5">300</td></tr>
<tr><td>Course Summary</td><td colspan="5">This course introduces students to the connection between mental health and physical fitness. It explains how physical activity can improve mental well-being and how mental health influences fitness and performance. Students will learn practical skills to design fitness programs that support mental health, including stress management and mindfulness exercises. By combining theory and practice, students will be prepared to promote both physical and mental health in sports and fitness settings.</td></tr>
<tr><td>Semester</td><td>6</td><td colspan="2">Credits</td><td>4</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course Details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>75</td></tr>
<tr><td>Pre-requisites, if any</td><td colspan="5">Students should have basic knowledge of human anatomy, exercise science, and general fitness principles.</td></tr>
</table>

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	**Explain how physical fitness and mental health are connected.**	U	1,2
CO2	Identify how exercise affects mood, stress, and mental well-being.	U	3
CO3	Plan fitness activities that help improve mental health.	S	2
CO4	Use simple tools to assess mental health and fitness levels.	C	4
CO5	Demonstrate practical mind-body exercises to reduce stress and promote wellness.	A	5,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	1	2	0	0	0	0	0	0	0	0
CO2	**0**	0	2	0	0	0	0	0	0	0
CO3	0	**1**	0	0	0	0	0	0	0	0
CO4	0	0	**0**	3	0	0	0	0	0	0
CO5	0	0	0	0	2	0	0	0	0	2
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Understanding Mental Health and Physical Fitness**		
	1.1	What is Mental Health, definition,	4	1
	1.2	What is Physical Fitness, components	4	2
	1.3	How Mental Health and Fitness Are Connected	4	3
	1.4	Common Mental Health Issues and Their Effects on Fitness	5	4
	1.5	Benefits of Physical Activity on Mental Health	5	2
2	2	**Psychological Factors and Physical Fitness**		
	2.1	Motivation to Exercise	4	3
	2.2	Stress and Anxiety: Causes and Effects	4	2
	2.3	Overcoming Barriers to Fitness	5	4
	2.4	Self-confidence and Exercise	5	2
	2.5	Exercise and Emotional Health	4	3
3	3	**Types of Exercise and Mental Health**		
	3.1	Aerobic Exercise and Mental Well-being	4	3
	3.2	Strength Training Benefits	4	4
	3.3	Yoga and Meditation Basics	3	2
	3.4	Group Exercise and Social Support	4	3
	3.5	Creating Balanced Fitness Plans	4	3
4	4	**Assessment and Safety in Mental Health & Fitness**		
	4.1	Simple Mental Health Screening Tools	2	3
	4.2	Physical Fitness Tests	2	2
	4.3	Adapting Exercise for Different Mental Health Conditions	2	4
	4.4	Safety and Ethics in Training	3	2
	4.5	When and How to Refer to Professionals	3	4
5		**TEACHER SPECIFIC CONTENT**		

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)**
	Mode of Assessment
	Classroom Procedure (Mode of transaction) ● Lecture (Power Point presentation) ● Demonstration ● Peer teaching ● Lesson plans ● Hands on training

Assessment Type	**MODE OF ASSESSMENT** **Continues Comprehensive Assessment (CCA)** **CCA for Theory:30 Marks** **Theory - CCA-15 mark, (MCQ 30x1=30)**
	End Semester Examination (ESE) Total Marks -70 **ESE Theory - (Duration of Examination 2 hr)** **Different part of examinations -** **Part 1, MCQ 10x1=10 (number of questions 12-number of questions to be answered 10)** **Part 2 - Short Answer – marks 5x2=10 (no. of questions 8 -number of questions to be answered 5) – Short Answer - Marks 5 x 4 =20 (no. of questions 6 -number of questions to be answered 4)Essay - Marks 15 x 2=30 (no. of questions 3 -number of questions to be answered 2)**

References

1. American College of Sports Medicine. (2018). ACSM's group exercise instructor manual (5th ed., pp. 215–235). Wolters Kluwer.
2. Bouchard, C., Blair, S. N., & Haskell, W. L. (2012). Physical activity and health (2nd ed., pp. 300–320). Human Kinetics.
3. Morgan, W. P., & O'Connor, P. J. (1988). Psychological bases of sport injuries and rehabilitation (pp. 95–115). Wiley.
4. Paluska, S. A., & Schwenk, T. L. (2000). Physical activity and mental health: Current concepts. Sports Medicine, 29(3), 167–180. https://doi.org/10.2165/00007256-200029030-00003
5. Ratey, J. J. (2008). Spark: The revolutionary new science of exercise and the brain (pp. 45–65). Little, Brown Spark.

Suggested Readings

1. Salmon, P. (2001). Effects of physical exercise on anxiety, depression, and sensitivity to stress: A unifying theory. Clinical Psychology Review, 21(1), 33–61. https://doi.org/10.1016/S0272-7358(99)00032-X
2. Segal, J., Smith, M., Robinson, L., & Segal, R. (2021). Exercise for mental health. HelpGuide.org. https://www.helpguide.org/articles/healthy-living/exercise-for-mental-health.htm
3. Vealey, R. S. (2007). Mental skills training in sport. In G. Tenenbaum & R. C. Eklund (Eds.), Handbook of sport psychology (3rd ed., pp. 287–309). Wiley.

Mahatma Gandhi University Kottayam

<table>
<tr><td>Faculty/ Discipline</td><td colspan="5">Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training</td></tr>
<tr><td>Programme</td><td colspan="5">B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training</td></tr>
<tr><td>Course Name</td><td colspan="5">Adventure Sports and Fitness</td></tr>
<tr><td>Type of Course</td><td colspan="5">SDE</td></tr>
<tr><td>Course Code</td><td colspan="5">MG6SDEMFT301</td></tr>
<tr><td>Course Level</td><td colspan="5">300</td></tr>
<tr><td>Course Summary</td><td colspan="5">This course explores the integration of adventure sports as a tool to enhance physical fitness. Students will engage in a variety of land, water, and aerial adventure activities that develop key fitness components such as strength, endurance, agility, and flexibility. The course also incorporates core strengthening techniques, risk management, safety, and team-based adventure planning. Through both theoretical and hands-on experience, students will gain a comprehensive understanding of how outdoor sports contribute to overall wellness and performance.</td></tr>
<tr><td>Semester</td><td>6</td><td colspan="2">Credits</td><td>4</td><td rowspan="2">Total Hours</td></tr>
<tr><td rowspan="2">Course Details</td><td rowspan="2">Learning Approach</td><td>Lecture</td><td>Practical</td><td>OJT</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>75</td></tr>
<tr><td>Pre-requisites, if any</td><td colspan="5">Basic understanding of physical fitness, general body conditioning, and exposure to outdoor fitness activities. Students must be physically fit for moderate to intense physical exertion.</td></tr>
</table>

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	**Explain the physiological and psychological benefits of adventure sports on fitness and well-being.**	**U**	1,2
CO2	Demonstrate core strengthening and endurance techniques through adventure-based activities.	**K**	**3,4**
CO3	Apply safety protocols and risk management practices in adventure sports environments.	**A**	**3,4**
CO4	Design and organize basic adventure fitness programs for individuals or groups.	**A**	**1,2,3**

CO5	Exhibit teamwork, communication, and leadership during outdoor and adventure training	**I**	**1,2,4**
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap			

CO-PO ARTICULATION MATRIX

CO/PO	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**
CO 1	**1**	**2**	**0**	0	0	0	0	0	0	0
CO2	**0**	0	2	1	0	0	0	0	0	0
CO3	**0**	0	1	3	0	0	0	0	0	0
CO4	1	2	**1**	0	0	0	0	0	0	0
CO5	1	1	0	3	0	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT

Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Fundamentals of Adventure Sports and Fitness**		
	1.1	Definition and scope of adventure sports	2	2
	1.2	Historical development and significance	2	1
	1.3	Types of adventure sports (Land, Water, Air)	2	1
	1.4	Benefits to physical and mental health	3	1
	1.5	Basic fitness components and their relevance to adventure activities	3	1
2	2	**Core Strengthening and Conditioning for Adventure Sports**		
	2.1	Anatomy of the core muscles and kinetic chains	4	2
	2.2	Functional training methods for adventure athletes	4	1
	2.3	Suspension, rope, and resistance training	5	1
	2.4	Natural terrain and bodyweight exercises	5	1
	2.5	Recovery and injury prevention techniques	4	1

3	3	**Adventure Sport Disciplines and Techniques**		
	3.1	Land-based activities: trekking, rock climbing, mountain biking	4	2
	3.2	Water-based activities: kayaking, rafting, open-water swimming	4	1
	3.3	Air-based activities: zip-lining, parasailing, paragliding (theoretical overview)	3	2
	3.4	Navigation and survival skills	4	2
	3.5	Equipment use and maintenance	4	3
4	4	**Safety, Risk Management & First Aid**		
	4.1	Risk factors in adventure environments	4	1
	4.2	Basic first aid and CPR training	4	2
	4.3	Emergency preparedness and wilderness survival	4	1
	4.4	Safety protocols and team briefing	5	1
	4.5	Legal and ethical issues in adventure sports	5	2
5		**TEACHER SPECIFIC CONTENT**		

<table>
<tr><td rowspan="3">Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)</td></tr>
<tr><td>Mode of Assessment</td></tr>
<tr><td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
● Lesson plans
● Group Activities</td></tr>
<tr><td rowspan="2">Mode of Assessment</td><td>Continues Comprehensive Assessment (CCA)
Total Mark - 30
<table>
<tr><th>Assessment Method (CCA)</th><th>Marks</th></tr>
<tr><td>Assignment</td><td>15</td></tr>
<tr><td>Individual Involvement & Viva</td><td>15</td></tr>
<tr><td>Total</td><td>30</td></tr>
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<tr><td>End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 70
Duration – 3 hrs
<table>
<tr><th>Assessment Method (ESE)</th><th>Marks</th></tr>
<tr><td>Lesson Plan</td><td>30</td></tr>
<tr><td>Demonstration & Presentation</td><td>30</td></tr>
<tr><td>Viva</td><td>10</td></tr>
<tr><td>Total</td><td>70</td></tr>
</table></td></tr>
</table>

References

1. Burchfield, R., & Patton, D. (2020). Adventure education: Theory and applications (pp. 85–105). Human Kinetics.
2. Fullagar, H. H. K., McCall, A., Impellizzeri, F. M., Favero, T. G., & Coutts, A. J. (2022). Load monitoring in adventure and outdoor sports (pp. 110–130). Routledge.
3. Heyward, V. H., & Gibson, A. L. (2022). Advanced fitness assessment and exercise prescription (8th ed., pp. 210–230). Human Kinetics.
4. Lewis, A. (2023). The outdoor fitness handbook: The ultimate guide for personal trainers (pp. 145–165). Bloomsbury Sport.
5. Meyers, M. C., & Laurent, C. M. (2021). Outdoor fitness: Training for performance in nature (pp. 100–120). Human Kinetics.

Suggested Readings

1. Norton, K., & Olds, T. (2023). Anthropometrica: A textbook of body measurement for sports and health (pp. 240–260). UNSW Press.
2. Sheel, A. W. (2023). Physiology of extreme environments and outdoor activities (pp. 175–195). Springer.
3. Wilson, J., & Codrington, J. (2021). Risk management in outdoor and adventure programs (2nd ed., pp. 190–210). Sagamore-Venture.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Advanced Training Designs & Periodization**
Type of Course	**SDC**
Course Code	**MG6SDCMFT300**
Course Level	300
Course Summary	This course introduces students to the advanced principles of training program design and periodization, which are essential for developing athletes and fitness clients. Students will explore the concepts of training cycles, adaptation mechanisms, performance peaking, and recovery. Emphasis will be placed on structuring training for sports, general fitness, and long-term athlete development, giving students practical tools to plan and evaluate training plans. The course bridges the gap between theory and application, preparing students to design effective, progressive training programs that align with performance goals.

Semester	6	**Credits**		4	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	1	0	75
Pre-requisites, if any	Basic understanding of human physiology and fitness components, Familiarity with general exercise science or fitness training principles, Interest in sports coaching or fitness program planning, Ability to engage in case study and data-based learning				

COURSE OUTCOMES (CO)

CONo	EXPECTED COURSE OUTCOMES (CO)	Learning Domains	PO No
CO1	**Understand and apply the core principles of training design, progression, and adaptation.**	**U**	1
CO2	Explain different periodization models and their relevance to performance training.	**U**	**4**

CO3	Design structured training plans across short-term and long-term cycles.	**U**	**2,1**
CO4	Evaluate training outcomes and make data-informed adjustments.	**E**	**5**
CO5	Integrate training variables such as intensity, volume, recovery, and tapering for athlete development.	**A**	**2**
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	1	0	0	0	0	0	0	0	0	0
CO2	0	0	0	1	0	0	0	0	0	0
CO3	1	**2**	0	0	0	0	0	0	0	0
CO4	0	0	0	0	2	0	0	0	0	0
CO5	0	1	0	0	0	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Principles of Training and Adaptation**		
	1.1	Principles of training: overload, specificity, progression, reversibility	5	1
	1.2	Physiological basis of adaptation – muscular, neural, metabolic	5	2
	1.3	Training variables: intensity, volume, frequency, density	5	2
	1.4	Acute vs. chronic responses to training stimulus	5	2
2	2	**Periodization Concepts and Models**		
	2.1	History and evolution of periodization	5	2
	2.2	Classic linear vs. non-linear (undulating) periodization	5	3
	2.3	Block periodization and conjugate methods	5	2

	2.4	Planning macrocycles, mesocycles, and microcycles	5	4
3	3	**Designing Training Plans for Performance**		
	3.1	Needs analysis for sport and fitness goals	4	2
	3.2	Structuring warm-up, main set, and cool-down in a program	4	1
	3.3	Integration of strength, endurance, agility, and flexibility	4	2
	3.4	Monitoring progress and adjusting workloads	4	1
4	4	**Practical Applications and Case Studies**		
	4.1	Case studies – Training plans for athletes vs. general clients	3	4
	4.2	Peaking and tapering strategies for competition	4	5
	4.3	Deloading and recovery week planning	4	2
	4.4	Common errors in training periodization and their solutions	4	5
		Teachers specific content		

<table>
<tr><td rowspan="3">Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)</td></tr>
<tr><td>Mode of Assessment</td></tr>
<tr><td>MODE OF ASSESSMENT Continuous Comprehensive Assessment (CCA) 25 Theory
CCA -15 marks (Assignment)
CCA -15 mark, (Presentation, Viva, individual involvement)</td></tr>
<tr><td>Mode of Assessment</td><td>Continues Comprehensive Assessment (CCA)
Total Mark - 30
<table>
<tr><th>Assessment Method (CCA)</th><th>Marks</th></tr>
<tr><td>Assignment</td><td>15</td></tr>
<tr><td>Individual Involvement & Viva</td><td>15</td></tr>
<tr><td>Total</td><td>30</td></tr>
</table>
End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 70
Duration – 3 hrs
<table>
<tr><th>Assessment Method (ESE)</th><th>Marks</th></tr>
<tr><td>Lesson Plan</td><td>30</td></tr>
<tr><td>Demonstration & Presentation</td><td>30</td></tr>
<tr><td>Viva</td><td>10</td></tr>
<tr><td>Total</td><td>70</td></tr>
</table></td></tr>
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References

1. Bompa, T. O., & Buzzichelli, C. (2019). Periodization: Theory and methodology of training (6th ed., pp. 110–130). Human Kinetics.
2. Hackett, D. A. (Ed.). (2023). Advanced resistance training: Techniques for performance (pp. 145–165). Routledge.
3. Issurin, V. B. (2021). Block periodization: Breakthrough in sport training (2nd ed., pp. 190–210). Ultimate Athlete Concepts.
4. Lorenz, D., & Morrison, S. (2022). Optimizing adaptation and performance: A training guide to periodization (2nd ed., pp. 160–180). Human Kinetics.
5. Mann, J. B. (2022). Developing explosive athletes: Use of velocity based training (pp. 95–115). JB Learning.

Suggested Readings

1. Naclerio, F., Jimenez, A., & Bishop, D. (2023). Periodization of strength training for sports and health (pp. 130–150). Springer.
2. Plisk, S. S., & Stone, M. H. (2020). Science and practice of strength training for sports (pp. 100–120). Routledge.
3. Stone, M. H., Stone, M., & Sands, W. A. (2023). Principles and practice of resistance training (2nd ed., pp. 175–195). Human Kinetics.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**				
Programme	**B.Voc .(Honours) Advanced Course in Multi Sports and Fitness Training**				
Course Name	**Officiating and Refereeing in Combat Sports (Judo/Taekwondo/Karate)**				
Type of Course	**SEC**				
Course Code	**MG6SECMFT300**				
Course Level	300				
Course Summary	This course introduces students to the principles, practices, and ethics of sports officiating in combat sports (Judo, Taekwondo, Karate). Students will learn rules, scoring, referee duties, and signaling systems, along with hands-on practice in officiating drills, mock competitions, and video analysis. The course also covers tournament organization, officiating professionalism, and safety protocols, preparing students for national- and state-level refereeing certifications.				
Semester	6	**Credits**		3	**Total Hours**
Course Details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	0	0	45
Pre-requisites, if any	Successful completion of introductory martial arts/combative sports paper - Basic knowledge of martial arts skills (Judo/Taekwondo/Karate). - Participation in at least one intercollegiate or university-level martial arts tournament (preferred).				

COURSE OUTCOMES (CO)

CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	**Understand the fundamental rules, regulations, and scoring systems of Judo, Taekwondo, and Karate**	**U**	1
CO2	Demonstrate practical officiating skills and apply correct referee signals in competitions.	**U**	**3**
CO3	Analyze competition structures, draw formats, and tournament organization in combat sports.	**C**	**2**

CO4	Develop decision-making, conflict-resolution, and communication skills required for refereeing.	**A**	**2, 5**
CO5	Apply knowledge of ethics, safety, and fair play in officiating martial arts events.Course Summary	**U**	**2**
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	1	0	0	0	0	0	0	0	0	0
CO2	0	0	2	0	0	0	0	0	0	0
CO3	0	**2**	0	0	0	0	0	0	0	0
CO4	0	1	0	0	2	0	0	0	0	0
CO5	0	1	0	0	0	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Introduction to Officiating**		
	1.1	History and development of officiating in Judo, Taekwondo, and Karate	5	1
	1.2	Roles, responsibilities, and qualities of referees and judges	4	2
	1.3	Basic tournament structures (round robin, knockout, repechage)	4	2
	1.4	Ethics, fair play, and code of conduct for officials	4	2
2	2	**Rules and Regulations**		
	2.1	Competition area specifications and safety guidelines	4	2
	2.2	Weight categories, bout duration, and competition attire	3	3
	2.3	Scoring system in Judo (Ippon, Waza-ari, penalties), Taekwondo (points, electronic scoring), Karate (ippon, waza-ari, yuko, penalties)	3	2

	2.4	Disqualifications, fouls, injury procedures	5	4
3	3	**Refereeing Signals and Practical Application**		
	3.1	Referee gestures and verbal commands (per sport)	4	2
	3.2	Duties of referees, judges, and table officials	3	1
	3.3	Practical simulation – officiating practice in class and mock bouts	3	2
	3.4	Use of technology – video replay systems, electronic scoring equipment	3	1
4	4	**TEACHER SPECIFIC CONTENT**		

<table>
<tr><td rowspan="3">Teaching and Learning Approach</td><td>Classroom Procedure (Mode of transaction)
● Lecture (Power Point presentation)
● Demonstration
● Peer teaching
● Lesson plans
● Hands on training</td></tr>
<tr><td>Mode of Assessment</td></tr>
<tr><td>Continues Comprehensive Assessment (CCA)
Total Mark - 25

<table>
<tr><th>Assessment Method (CCA)</th><th>Marks</th></tr>
<tr><td>Assignment</td><td>10</td></tr>
<tr><td>Individual Involvement & Viva</td><td>15</td></tr>
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<tr><td>Assessment Type</td><td>End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 50
Duration – 2 Hrs

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<tr><td>Viva</td><td>10</td></tr>
<tr><td>Total</td><td>50</td></tr>
</table></td></tr>
</table>

References (APA 7 Style)

1. International Judo Federation. (2023). IJF refereeing rules and guidelines. IJF.
2. Kukkiwon. (2022). Official Taekwondo competition rules. Kukkiwon Publication.
3. World Karate Federation. (2023). WKF competition rules. WKF.
4. Singh, A., & Sharma, R. (2019). Martial arts: Techniques, training and officiating. New Delhi: Sports Publication.
5. Balyi, I., Way, R., & Higgs, C. (2013). Long-term athlete development. Human Kinetics.

Suggested Readings

1. Jones, R. L., & Kingston, K. (2013). An introduction to sports coaching: Connecting theory to practice. Routledge.
2. Weinberg, R. S., & Gould, D. (2018). Foundations of sport and exercise psychology (7th ed.). Human Kinetics.
3. United States Olympic & Paralympic Committee. (2020). Coaching and officiating education manual. USOPC.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Fitness Centre Management**
Type of Course	**MPC**
Course Code	**MG6MPCMFT300**
Course Level	300
Course Summary	This course introduces students to the practical and managerial aspects of running a fitness centre. It teaches how to plan, organize, and manage daily operations in a gym or health club setting. Students will learn about facility setup, staff roles, client safety, marketing, finance, and legal responsibilities. The course also emphasizes safety precautions, hygiene, and customer service to ensure smooth and professional management of fitness facilities.

Semester	6	**Credits**		4	**Total Hours**
Course details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		4	0	0	**60**
Pre-requisites, if any	Students must have basic understanding of fitness activities, and an interest in managing or working in the fitness industry. No prior management or business experience required				

COURSE OUTCOMES (CO)

CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	**Identify safety protocols, hygiene practices, and legal guidelines in fitness centers.**	U	2
CO2	Understand the structure and functions of a fitness center.	U	2
CO3	Apply basic management principles in operating a fitness facility.	U	2
CO4	Plan client services, staff schedules, and equipment management.	K	2,3
CO5	Create simple business, marketing, and customer service strategies for a gym.	C	4
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap			

CO-PO ARTICULATION MATRIX

CO/PO	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**
CO1	0	2	0	0	0	0	0	0	0	0
CO2	0	3	0	0	0	0	0	0	0	0

CO3	**0**	2	0	0	0	0	0	0	0	0
CO4	0	1	3	0	0	0	0	0	0	0
CO5	0	0	0	3	0	0	0	0	0	0
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	**Introduction to Fitness Center Operations**		
	1.1	Types of fitness centers and services offered	3	1
	1.2	Organizational structure and staff roles	3	1
	1.3	Layout and infrastructure planning	3	1
	1.4	Equipment selection and basic maintenance	4	2
	1.5	Facility rules, membership policies, and client forms	4	1
2	2	**Health, Safety, and Hygiene Management**		
	2.1	Basic health screening for clients	3	2
	2.2	Cleanliness, sanitization, and infection control	3	3
	2.3	Emergency protocols and first aid preparedness	4	2
	2.4	Injury prevention and safe equipment use	4	2
	2.5	Legal liability and risk management	3	3
3	3	**Human Resource & Staff Management**		
	3.1	Roles and responsibilities of trainers and staff	3	3
	3.2	Staff recruitment, training, and scheduling	3	2
	3.3	Communication skills and professionalism	2	5
	3.4	Conflict resolution and team coordination	3	2
	3.5	Code of conduct and ethics in fitness settings	3	1
4	4	**Client Services and Customer Management**		
	4.1	Client onboarding and assessment process	2	2
	4.2	Designing personalized fitness plans	2	3
	4.3	Client tracking and progress evaluation	2	4
	4.4	Managing client feedback and grievances	3	2
	4.5	Creating a welcoming and motivational environment	3	4
5		**TEACHER SPECIFIC CONTENT**		

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)** **● Lecture (Power Point presentation)** **● Demonstration** **● Peer teaching** **● Lesson plans** **● Hands on training**
	Mode of Assessment
	MODE OF ASSESSMENT Continuous Comprehensive Assessment (CCA) **CCA for Theory:30 Marks** CCA -15 marks (Assignment, viva) CCA -15 mark, (Presentation, individual involvement)
Assessment Type	**End Semester Examination (ESE) -** **ESE Theory –70 marks** (**Written examination theory) -** **Duration of Examination 2 hrs -** **Different part of examinations** – **Part 1**, MCQ 10x1=10 (number of questions 12-number of questions to be answered 10) **Part 2** - Short Answer – marks - 5x2=10 (number of questions 7 -number of questions to be answered 5) **Part 3 -** Short Essay - 4x5=20 (number of questions 6-number of questions to be answered 4) **Part 4** - Essay-2 x 15=30 - (number of questions 3-number of questions to be answered 2).

References

1. Brawley, L. R. (2023). Fitness management: A strategic approach (pp. 85–100). Routledge.
2. Bubbico, R. (2021). Fitness business book: Strategies for success in the gym industry (pp. 140–155). Fitness Prof Press.
3. Fallon, L. (2022). Facility management for physical activity and sport (pp. 190–205). Routledge.
4. Howley, E. T., & Thompson, D. L. (2023). Fitness professional's handbook (8th ed., pp. 210–225). Human Kinetics.
5. Kennedy-Armbruster, C., & Yoke, M. M. (2022). Methods of group exercise instruction (4th ed., pp. 160–175). Human Kinetics.

Suggested Readings

1. Robinson, L. (2021). Sport club and facility management (pp. 300–315). Cengage Learning.
2. Weir, J. P. (2022). Essentials of strength training and conditioning (4th ed., pp. 280–295). Human Kinetics.
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4. Yoke, M. M. (2021). Client-centered fitness management (pp. 115–130). Jones & Bartlett Learning.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences / Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**Basics of Geriatric Fitness**
Type of Course	**VAC**
Course Code	**MG6VACMFT300**
Course Level	300
Course Summary	This course introduces students to the foundational concepts and hands-on skills required to assess, design, and implement safe and effective fitness programs for elderly populations. Students will engage in practical sessions aimed at promoting physical independence, mobility, strength, and balance in older adults, considering age-related physiological and psychological changes.

Semester	6	**Credits**		3	**Total Hours**
Course details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		3	0	0	**45**
Pre-requisites, if any	Basic knowledge of human anatomy, exercise science, and general fitness training principles covered in the first-year courses of the fitness training program.				

COURSE OUTCOMES (CO)

CONo	**EXPECTED COURSE OUTCOMES (CO)**	**Learning Domains**	**PO No**
CO1	Identify the physiological changes associated with aging and their impact on physical performance.	U	1,2,10
CO2	Perform appropriate fitness assessments and screenings for older adults.	U	2,4,5
CO3	Demonstrate safe and effective exercises for elderly individuals with varying health conditions.	AN	2,4
CO4	Design age-appropriate fitness routines for improving strength, flexibility, balance, and endurance.	U	2,4,5
CO5	Apply injury prevention, fall risk reduction, and safety protocols in geriatric exercise settings.	A	2,4,10

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap

CO-PO ARTICULATION MATRIX										
CO/PO	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**
CO1	1	1	0	0	0	0	0	0	0	2
CO2	**0**	2	0	1	1	0	**0**	0	0	0
CO3	0	**2**	0	2	**0**	**0**	0	0	**0**	0
CO4	0	3	**0**	1	2	0	0	0	0	0
CO5	0	1	0	1	0	0	0	0	0	1
'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).										

COURSE CONTENT				
Module	**Units**	**Course description**	**Hrs**	**CO No.**
1	1	Understanding Aging and Fitness		
	1.1	Introduction to Aging and Geriatric Fitness	4	3
	1.2	Physiological Changes in Musculoskeletal, Cardiovascular, and Nervous Systems	4	2
	1.3	Common Chronic Conditions in the Elderly (Arthritis, Osteoporosis, Diabetes, etc.)	3	1
	1.4	Benefits of Physical Activity in Old Age	4	2
2	2	Assessment and Screening Techniques		
	2.1	Pre-screening Tools and Health History Forms (PAR-Q+)	4	2
	2.2	Functional Fitness Tests (Chair Stand, Arm Curl, 6-min Walk, etc.)	3	1
	2.3	Balance, Mobility, and Fall Risk Assessments	4	2
	2.4	Psychological and Cognitive Fitness Assessment Tools	4	3
3	3	Practical Exercise Programming		
	3.1	Strength and Resistance Training Using Bands and Bodyweight	4	4
	3.2	Flexibility and Range of Motion Exercises	3	3
	3.3	Balance, Coordination, and Fall Prevention Techniques	4	2
	3.4	Low-Impact Aerobic Workouts for Elderly Clients	4	5

4		**TEACHER SPECIFIC CONTENT**		

Teaching and Learning Approach	**Classroom Procedure (Mode of transaction)**
	Mode of Assessment
	Continues Comprehensive Assessment (CCA) Total Mark - 25 Marks

Assessment Method (CCA)	Marks
Assignment	10
Individual Involvement & Viva	15
Total	25

Assessment Type

End Semester Examination (ESE)
Assessment Method – Practical Examination
Total Marks – 50

Assessment Method (ESE)	Marks
Lesson Plan	20
Demonstration & Presentation	20
Viva	10
Total	50

References

1. Chodzko-Zajko, W., Proctor, D. N., Fiatarone Singh, M. A., Minson, C. T., Nigg, C. R., Salem, G. J., & Skinner, J. S. (2019). *Exercise and physical activity for older adults*. Human Kinetics. **(pp. 108–132)**
2. Frontera, W. R., & Slovik, D. M. (2023). *Clinical exercise physiology for aging adults* (3rd ed.). Elsevier. **(pp. 87–150)**
3. Nelson, M. E., & Rejeski, W. J. (2021). *Physical activity and public health in older adults*. American College of Sports Medicine (ACSM). **(pp. 94–120)**
4. Hunter, G. R., McCarthy, J. P., & Bamman, M. M. (2021). *Resistance training for older adults*. Human Kinetics. **(pp. 33–95)**
5. Seguin, R. A., & Nelson, M. E. (2022). *Exercise for the older adult* (2nd ed.). American College of Sports Medicine (ACSM). **(pp. 65–112)**

Suggested Readings

1. Mazzeo, R. S. (2020). *ACSM's guide to exercise and aging*. Lippincott Williams & Wilkins. **(pp. 52–118)**
2. Evans, W. J., & Rosenberg, I. H. (2022). *Biomarkers of aging and exercise interventions*. Springer. **(pp. 102–145)**
3. King, A. C., & King, D. K. (2023). *Active living and public health for seniors*. Springer. **(pp. 70–130)**

Mahatma Gandhi University Kottayam

Faculty	**Physical Education and Sport Sciences/Advanced Course in Multi Sports and Fitness Training**				
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**				
Course Name	**Project**				
Type of Course	**PRJ**				
Course Code	**MG6PRJMFT300**				
Course Summary	The project work provides students with an opportunity to identify, analyze, and solve real-world problems relevant to their field of study by integrating and applying the theoretical knowledge and skills acquired throughout their academic program. It fosters independent research, critical thinking, innovation, and the practical use of methodologies, tools, and techniques to design effective solutions. Students are encouraged to work individually or in teams, enhancing their collaboration, time management, ethical responsibility, and self-directed learning. The project also develops competencies in academic writing, documentation, and technical communication. Each project is expected to culminate in a comprehensive report, a working model or prototype (where applicable), and a formal presentation followed by a viva voce examination, demonstrating the student's ability to apply knowledge creatively and professionally in a real-world context.				
Semester	6	**Credits**		4	
Course details	**Learning Approach**	**Lecture**	**Practical**	**OJT**	
		0	8	0	

COURSE OUTCOME (CO)

	Upon the successful completion of the course, the student will be able to		
	Identify, analyze, and define problems relevant to the field of study.	An	1,2,3
	Apply appropriate methodologies, tools, and techniques to design and implement effective solutions.	C	2,3,10
	Demonstrate skills in research, critical thinking, project planning, and systematic execution.	S	1,2,5,10
	Produce well-structured academic reports and communicate project outcomes effectively.	S	4,8,10
	Exhibit teamwork, time management, ethical responsibility, and initiative in a self-directed project environment.	S	5,8,9,10

	Address real-world challenges with innovative and context-aware solutions.	Ap	1,2,6,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	3	1	0	0	0	0	0	0	0
CO 2	0	3	2	0	0	0	0	0	0	2
CO 3	3	3	0	0	2	0	0	0	0	3
CO 4	0	0	0	2	0	0	0	3	0	3
CO 5	0	0	0	0	3	0	0	2	1	1
CO 6	3	2	0	0	0	1	0	0	0	1

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

Assessment Types	**MODE OF ASSESSMENT**		
	A	**Continuous Comprehensive Assessment(CCA)**	
		Components	**Marks**
		Commitment and Involvement	5
		Periodic progress review	10
		Quality of work/Implementation effort	10
		Report	5
		Total	**30**
	B	**End Semester Evaluation(ESE)**	
		Components	**Marks**
		Problem Identification and Objectives	10
		Methodology / Design / Technical Content	15
		Implementation / Analysis / Results	15
		Final Report	10
		Presentation	10
		Viva Voce	10
		Total	**70**

SEMESTER VII & VIII

B.VOC HONOURS							
COURSE CODE	TITLE OF THE COURSE	TYPE OF THE COURSE	CREDIT	NO.OF DAYS	CREDIT DISTRIBUTION		
					L	P	O
MG7APPMFT400	APPRENTICESHIP	APP	28	280	0	28	0
	Online	MPC	4				
	Online	MPC	4				
	Online	MPC	4				

BVOC HONOURS WITH RESEARCH							
COURSE CODE	TITLE OF THE COURSE	TYPE OF THE COURSE	CREDIT	NO.OF DAYS	CREDIT DISTRIBUTION		
					L	P	O
MG7RINMFT400	RESEARCH INTERNSHIP	RIN	20	200	0	20	0
	Online	SDC	4				
	Online	SDC	4				
	Online	MPC	4				
	Online	MPC	4				
	Online	MPC	4				

L — Lecture, P — Practical/Practicum , O — On-the-Job Training

	Mahatma Gandhi University **Kottayam**

Faculty/ Discipline	**Physical Education and Sport Sciences/Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course In Multi Sports And Fitness Training**
Course Name	**APPERTENCISHIP**
Course Code	**MG7APPMFT400**
Course Level	400
Course Summary	As an integral component of the B.Voc. Honours degree programme, students are required to complete a structured apprenticeship or work-integrated learning programme in collaboration with relevant industries, organizations, or institutions. This component, spanning a duration of 280 days, carries 28 academic credits and is compulsory in the student's designated skill domain. It is designed to enhance industry preparedness by reinforcing academic knowledge through sustained, domain-relevant practical experience. The apprenticeship offers students the opportunity to engage directly with real- world professional environments, enabling them to apply domain-specific competencies, gain exposure to industry-standard tools and practices, and participate meaningfully in ongoing operations and projects. This extended, immersive experience serves to bridge the gap between theoretical learning and professional expectations, thereby fostering critical skills for career development and employability. To ensure the effectiveness, academic relevance, and accountability of the apprenticeship: • Each student will be assigned an academic mentor from the parent institution and an industry supervisor from the host organization. • Students are required to maintain a weekly activity logbook, which must be regularly reviewed and signed by the industry supervisor. • Monthly progress reports will be submitted to and reviewed by the academic mentor in consultation with the industry supervisor.

	• Mid-term and final evaluations will be conducted based on a combination of employer feedback, student outputs/deliverables, and academic performance metrics. • The institution will conduct site visits, virtual check-ins, or regular follow-ups to ensure student engagement, address issues promptly, and uphold the quality of the apprenticeship experience. This structured apprenticeship is a critical step in preparing students for the dynamic demands of the professional world, ensuring that their academic journey culminates in a well-rounded and industry-aligned skill set.				
Semester	**7 & 8**	**Duration**	**280 days**	**Credits**	**28**

COURSE OUTCOMES (CO)

CO No:	Expected Course Outcome	Learning Domains	PO No:
	Upon the successful completion of the course, the student will be able to		
1	Gain hands-on professional experience by engaging in long-term, domain-specific apprenticeship in real- world industry environments.	S	1,3,6,10
2	Apply domain-specific theoretical knowledge to solve real-time problems, enhancing technical and problem-solving competencies.	A	1,2,3,10
3	Demonstrate professional competencies such as workplace etiquette, communication skills, and teamwork in a collaborative work culture.	S	4,5,8,9
4	Build a professional portfolio by achieving practical outcomes and establishing credible industry references and credentials.	C	5,9,10
5	Cultivate reflective thinking, adaptability, and a lifelong learning mindset through structured and mentored work experience.	Ap	1,6,8,10
6	Transition smoothly from academic study to professional practice by developing job-specific skills and industry-aligned competencies.	S	2,3,5,10

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Assessment Types	**MODE OF ASSESSMENT**		
	A	**Continuous Comprehensive Assessment(CCA)**	
		Components	**Marks**
		Commitment, Punctuality & Professional Conduct	10
		Monthly Progress Reviews & Logbook Maintenance	25
		Skill Development & Application	25
		Interim Report	20
		Total	**90**
	B	**End Semester Evaluation(ESE)**	
		Components	**Marks**
		Feedback & Evaluation Report from Host Organization	50
		Skill Demonstration / Summary of Work Exposure	40
		Final Report / Learning Portfolio	40
		Domain Knowledge and Experience Communication (Presentation)	40
		Viva Voce	40
		Total	**210**

Note:

This assessment framework is intended as a guiding structure for evaluating apprenticeship performance. However, in order to remain responsive to the evolving needs of industry and society, the evaluation criteria may be revised from time to time. Such changes aim to enhance the relevance, effectiveness, and fairness of the assessment process.

	Mahatma Gandhi University **Kottayam**

Faculty/ Discipline	**Physical Education and Sport Sciences/Advanced Course in Multi Sports and Fitness Training**
Programme	**B.VOC. (HONOURS) ADVANCED COURSE IN MULTI SPORTS AND FITNESS TRAINING**
Course Name	**RESEARCH INTERNSHIP**
Type of Course	**RIN**
Course Code	**MG7RINMFT400**
Course Level	400
Couse Summary	As an integral requirement of the B.Voc. Honours with Research degree programme, the Research Internship is designed to provide students with hands-on exposure to real-world research practices in their designated skill domain. This component carries 20 academic credits and extends over a duration of 200 days. The internship must be undertaken in collaboration with a research organization, industry, or university department, under the mentorship of a qualified research guide. The primary aim of this internship is to engage students in industry-linked research projects that allow them to apply theoretical knowledge to practical, domain-specific problems. Students are expected to work on meaningful research inquiries, contribute to data collection and analysis, develop critical thinking and problem-solving skills, and enhance their communication and documentation abilities. In addition to the research internship, students must earn 8 credits through Skill Development Courses (SDCs), specifically chosen for their research orientation, thereby reinforcing their academic and practical foundation. This component not only contributes significantly to the academic rigor of the Honours with Research degree but also ensures a seamless transition from classroom learning to workplace research, preparing students for advanced studies or professional roles in their respective domains.

Semester	**7&8**	**Duration**	**200 days**	**Credits**	**20**

COURSE OUTCOMES (CO)

CO No:	Expected Course Outcome	Learning Domains	PO No:
	Upon the successful completion of the course, the student will be able to		
1	Demonstrate research aptitude and inquiry-based learning by actively engaging in real-time research projects.	S	1,2,10
2	Apply academic knowledge in a professional research environment to bridge the gap between theory and real-world research practices.	A	2,3,6,10
3	Strengthen domain-specific knowledge and technical competencies through systematic investigation and practical application.	S	1,2,3
4	Address real-world research problems using problem-solving, analytical, and critical thinking skills.	S	1,2,6
5	Communicate scientific ideas and findings effectively through research reports, documentation, and presentations.	S	4,8,10
6	Collaborate with researchers and peer groups to gain exposure to interdisciplinary perspectives and collaborative learning practices.	S	2,3,5,10
7	Demonstrate professional growth and readiness for higher education, entrepreneurship, or research-oriented careers.	I	5,9,10
****Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)***			

O-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	0	0	0	0	0	0	0	2
CO 2	0	3	2	0	0	2	0	0	0	1
CO 3	3	2	3	0	0	0	0	0	0	0
CO 4	3	2	0	0	0	2	0	0	0	0
CO 5	0	0	0	3	0	0	0	2	0	1
CO 6	0	3	3	0	2	0	0	0	0	1
CO 7	0	0	0	0	3	0	0	0	2	1

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

Assessment Types	**MODE OF ASSESSMENT**		
	A	**Continuous Comprehensive Assessment(CCA)**	
		Components	**Marks**
		Commitment, Punctuality & Professional Conduct	10
		Monthly Progress Reviews & Logbook Maintenance	15
		Skill Development & Application	15
		Interim Report	20
		Total	**60**
	B	**End Semester Evaluation(ESE)**	
		Components	**Marks**
		Feedback & Evaluation Report from Host Organization	40
		Skill Demonstration / Summary of Work Exposure	20
		Final Report / Learning Portfolio	25
		Domain Knowledge and Experience Communication (Presentation)	25
		Viva Voce	30
		Total	**140**

Note:

This assessment framework serves as a guiding structure for evaluating research internship performance. However, to remain responsive to the evolving needs of industry, academia, and society, the evaluation criteria may be revised periodically. Such revisions aim to enhance the relevance, effectiveness, and fairness of the overall assessment process.

Mahatma Gandhi University Kottayam

Faculty/ Discipline	**Physical Education and Sport Sciences/Advanced Course in Multi Sports and Fitness Training**
Programme	**B.Voc. (Honours) Advanced Course in Multi Sports and Fitness Training**
Course Name	**On-the-Job Training**
Type of Course	**SDC**
Course Summary	On-the-Job Training (OJT) is designed to equip students with practical skills, workplace discipline, and industry exposure by actively engaging them in real-world professional environments. Conducted in collaboration with firms, industries, research institutions, or higher education establishments, OJT enables students to understand industry standards, apply academic knowledge, and perform job-specific tasks using contemporary tools and practices. The training must be undertaken in the student's own skill domain, aligned with the major area of study in their undergraduate program, to ensure relevance and coherence with their academic and career goals. The program also fosters essential workplace competencies such as communication, responsibility, adaptability, and teamwork. Furthermore, it offers students a platform for career exploration and networking, helping them evaluate potential career paths and align their aspirations with industry demands.

Semester	**1,2,3**	**Duration**	**5 hours/week**	**Credits**	**2**

COURSE OUTCOMES (CO)

CO No:	Expected Course Outcome	Learning Domains	PO No:
	Upon the successful completion of the course, the student will be able to		
1	Demonstrate understanding of industry operations, standards, and professional expectations through direct exposure to workplace environments.	Ap	1,3,6,10

2	Apply job-specific skills effectively in real-world tasks and responsibilities within the assigned industry setting.	S	2,4,5,10
3	Integrate academic knowledge with practical applications to solve work-related challenges and contribute to organizational goals.	An	1,2,3,6
4	Exhibit essential workplace competencies such as punctuality, accountability, communication, teamwork, and adaptability.	S	4,5,8,9
5	Identify and evaluate potential career opportunities by reflecting on their internship experiences and professional interactions.	E	1,9,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

<table>
<tr><td rowspan="12">Assessment Types</td><td colspan="3">MODE OF ASSESSMENT</td></tr>
<tr><td>A</td><td colspan="2">Continuous Comprehensive Assessment(CCA)</td></tr>
<tr><td rowspan="4"></td><td>Components</td><td>Marks</td></tr>
<tr><td>Feedback from the hosting organization</td><td>5</td></tr>
<tr><td>Internal Supervisor feedback</td><td>10</td></tr>
<tr><td>Total</td><td>15</td></tr>
<tr><td>B</td><td colspan="2">End Semester Evaluation(ESE)</td></tr>
<tr><td rowspan="5"></td><td>Components</td><td>Marks</td></tr>
<tr><td>Presentation</td><td>10</td></tr>
<tr><td>Report</td><td>10</td></tr>
<tr><td>Viva Voce</td><td>15</td></tr>
<tr><td>Total</td><td>35</td></tr>
</table>